
第三课 单细胞转录组计算方法与数据分析

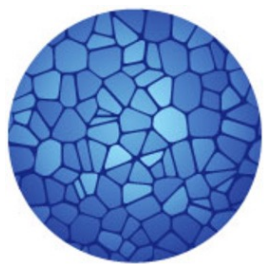
中国医学科学院基础医学研究所

陈阳

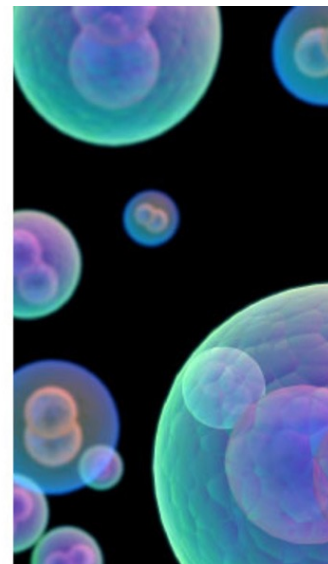
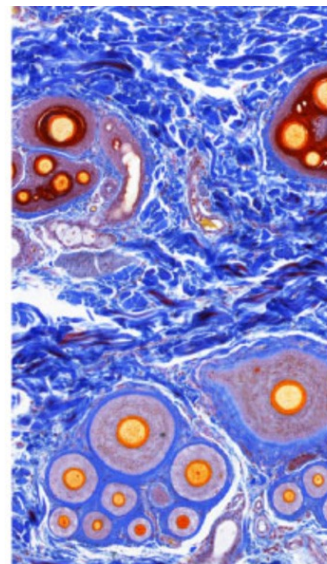
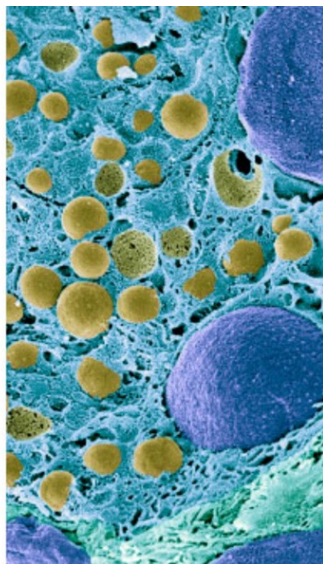
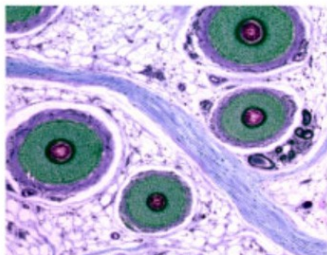
yc@ibms.pumc.edu.cn

20260416

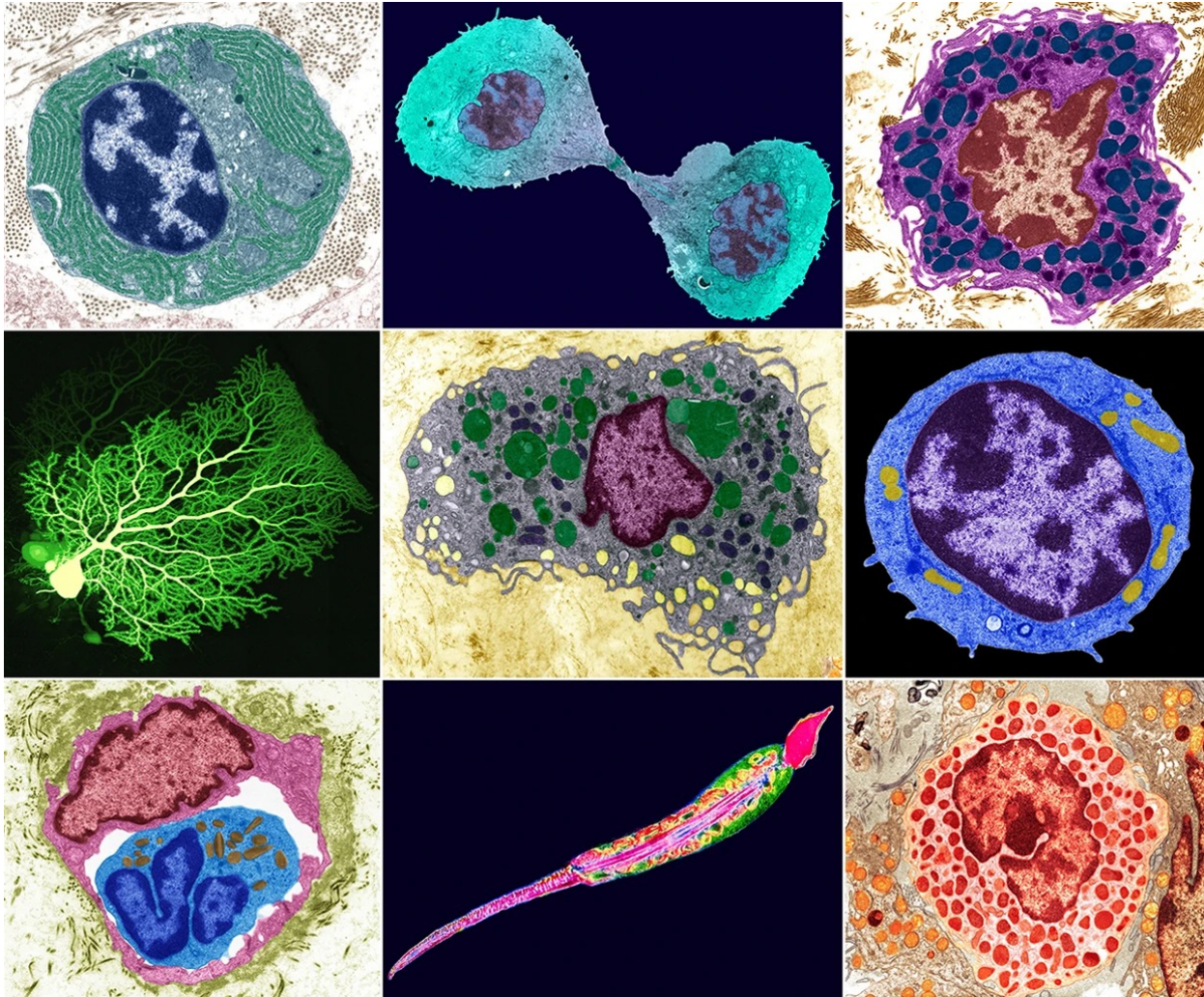
平均而言每个人由40-60万亿个细胞组成



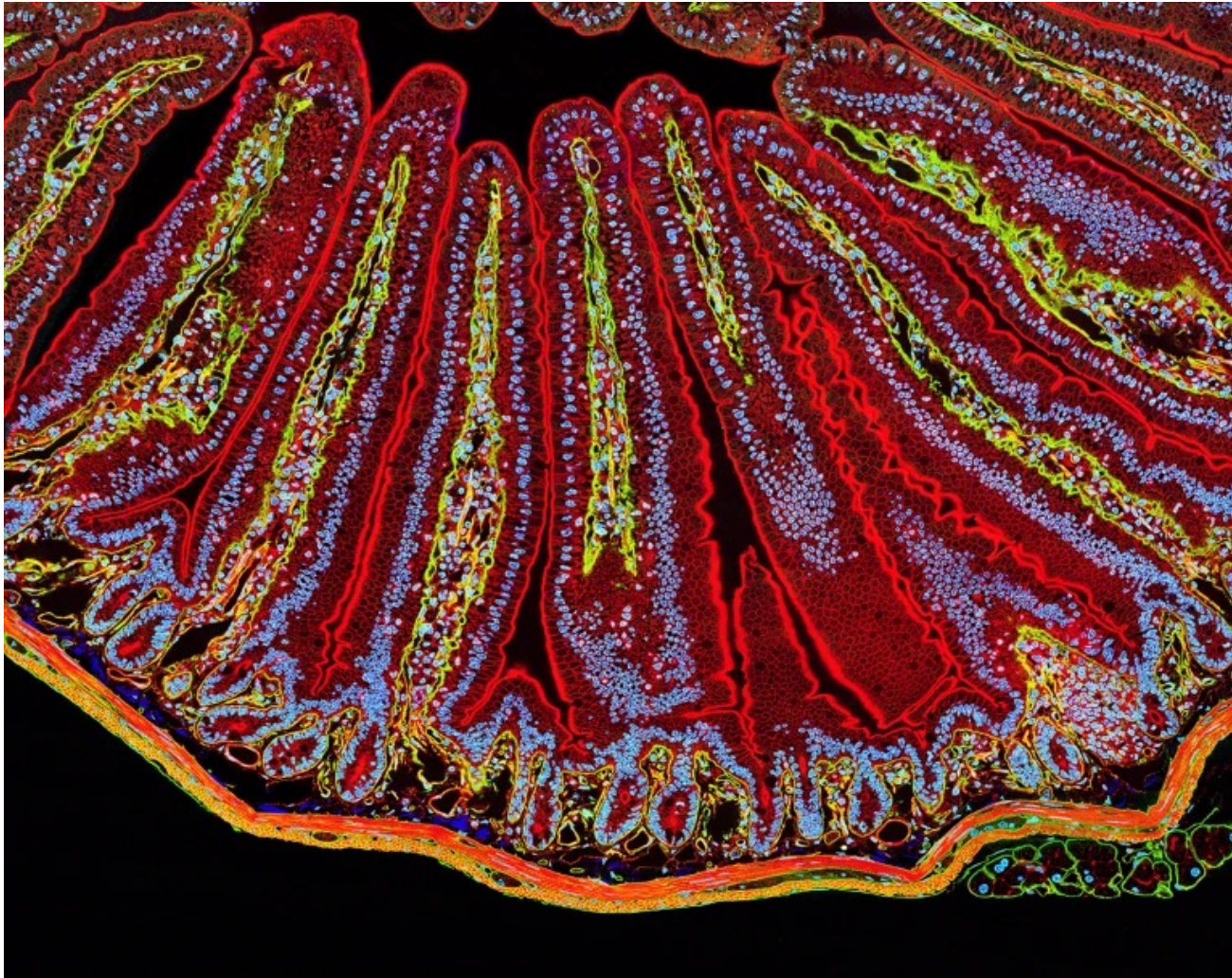
**HUMAN
CELL
ATLAS**



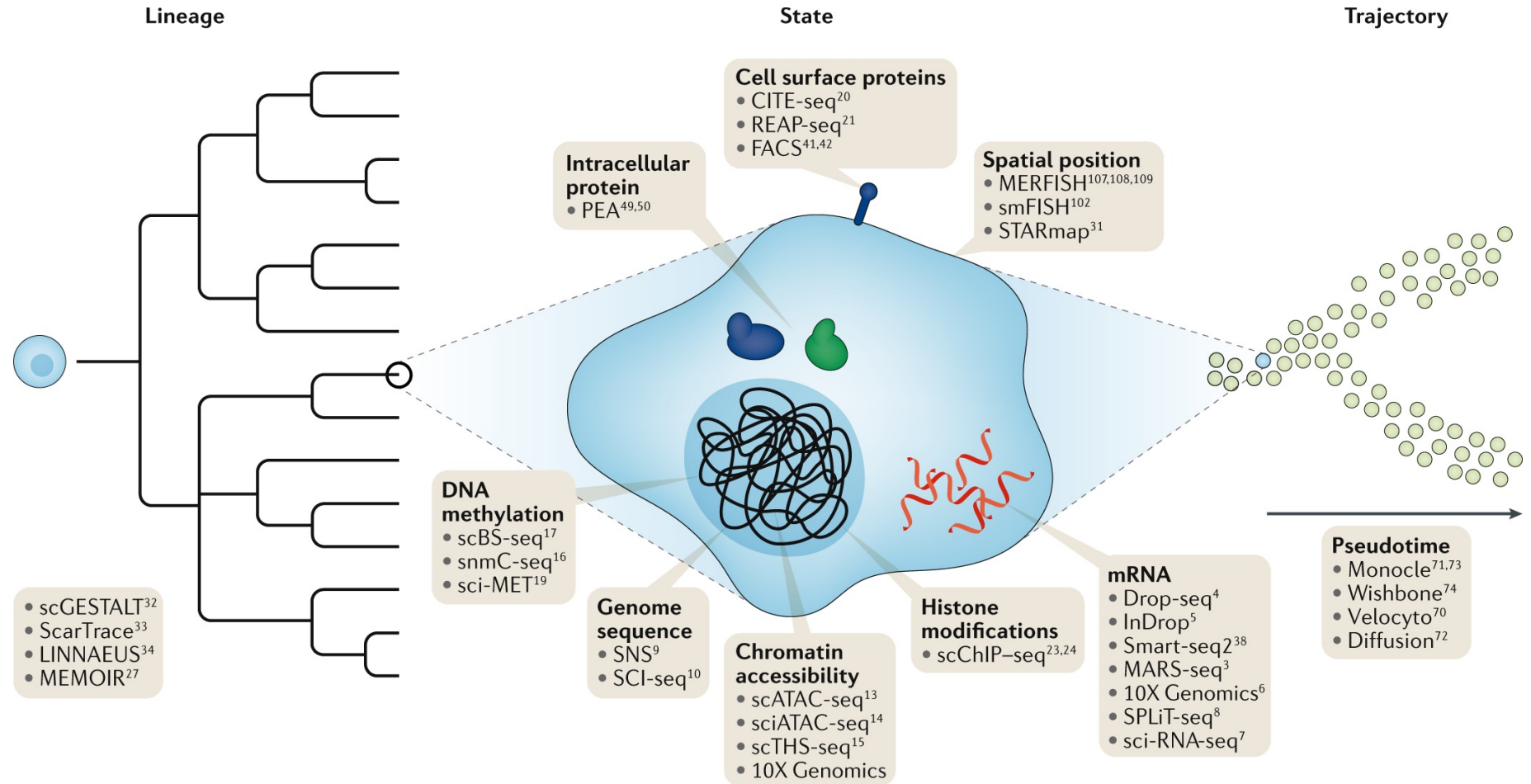
What is a cell type, really?



Villi in the small intestine

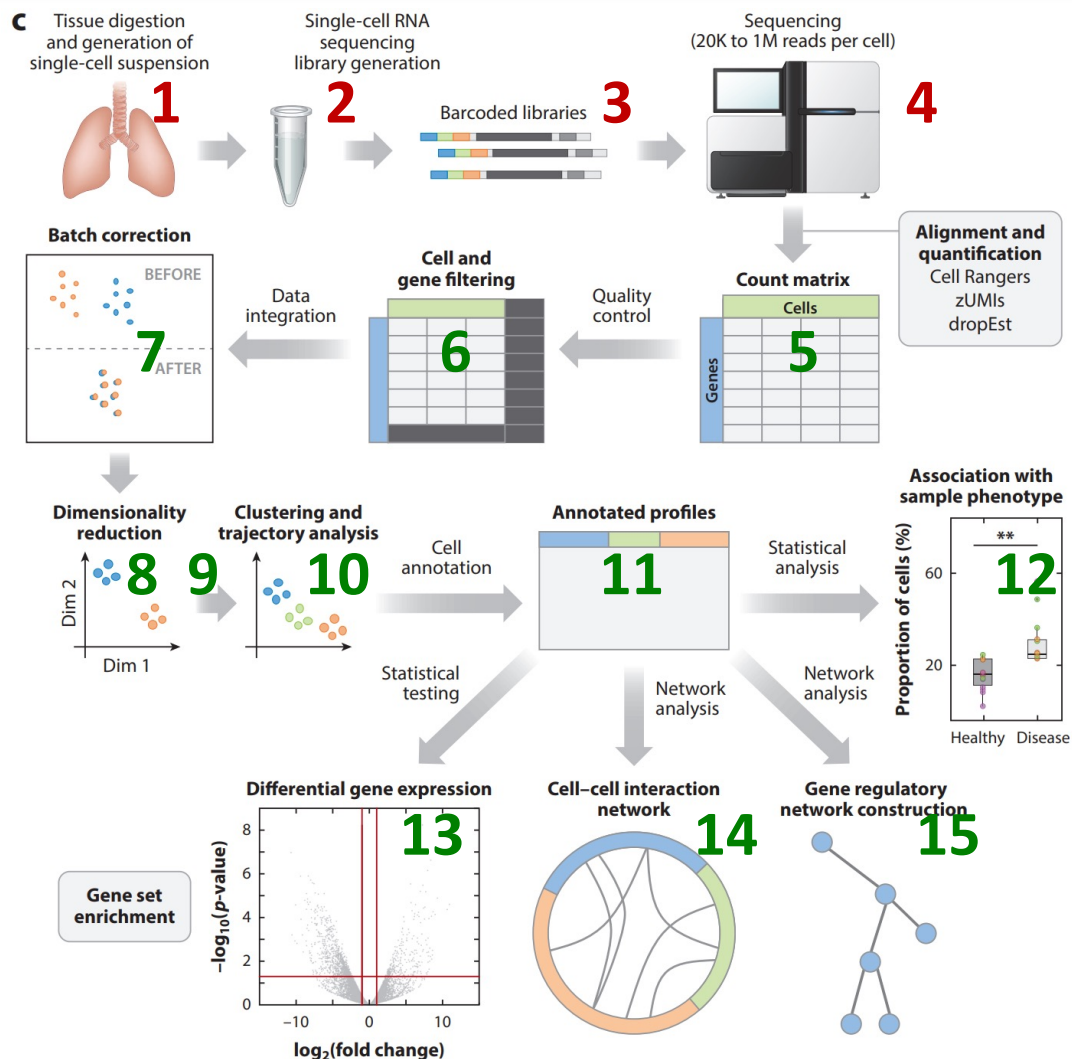


单细胞组简介



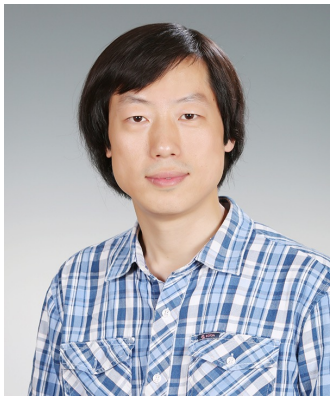
单细胞转录组技术及分析基本流程

- 1-样本
- 2-文库
- 3-文库
- 4-测序
- 5-比对
- 6-过滤
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- 8-归一化
- 9-降维
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- 14-通讯
- 15-调控



scRNA-seq

分子实验设计



汤富酬

nature methods

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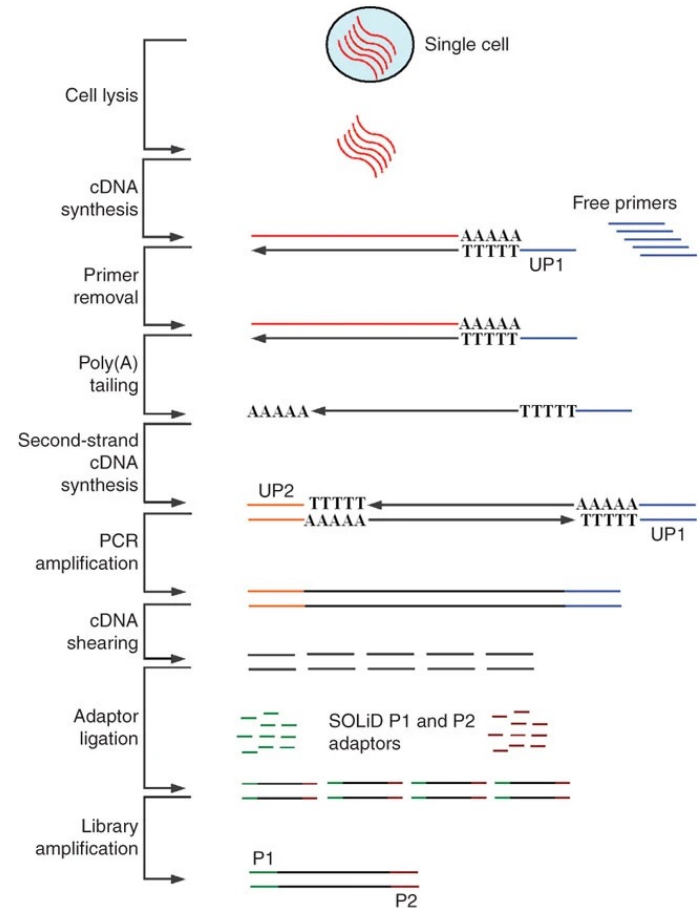
Published: 06 April 2009

mRNA-Seq whole-transcriptome analysis of a single cell

Fuchou Tang, Catalin Barbacioru, Yangzhou Wang, Ellen Nordman, Clarence Lee, Nanlan Xu, Xiaohui Wang, John Bodeau, Brian B Tuch, Asim Siddiqui, Kaiguo Lao & M Azim Surani

[Nature Methods](#) 6, 377–382 (2009) | [Cite this article](#)

47k Accesses | 1847 Citations | 126 Altmetric | [Metrics](#)



2009年，汤富酬进行第一个基于NGS的单细胞转录组测序

<https://www.nature.com/articles/nmeth.1315>

STRT-seq

分子实验设计



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Characterization of the single-cell transcriptional landscape by highly multiplex RNA-seq

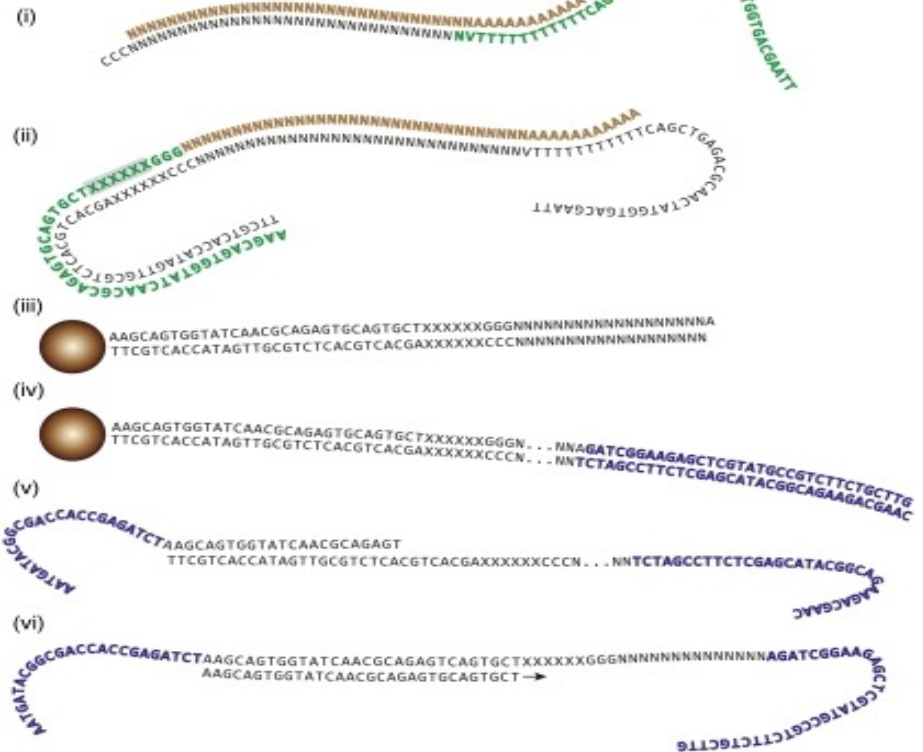
Saiful Islam^{1,4}, Una Kjällquist^{1,4}, Annalena Moliner², Pawel Zajac¹,
Jian-Bing Fan³, Peter Lönnerberg¹ and Sten Linnarsson^{1,5}



Sten Linnarsson
Karolinska Institute

2011年，首次引入barcode

A



SMART-seq

分子实验设计



Rickard Sandberg
Karolinska Institutet

nature biotechnology

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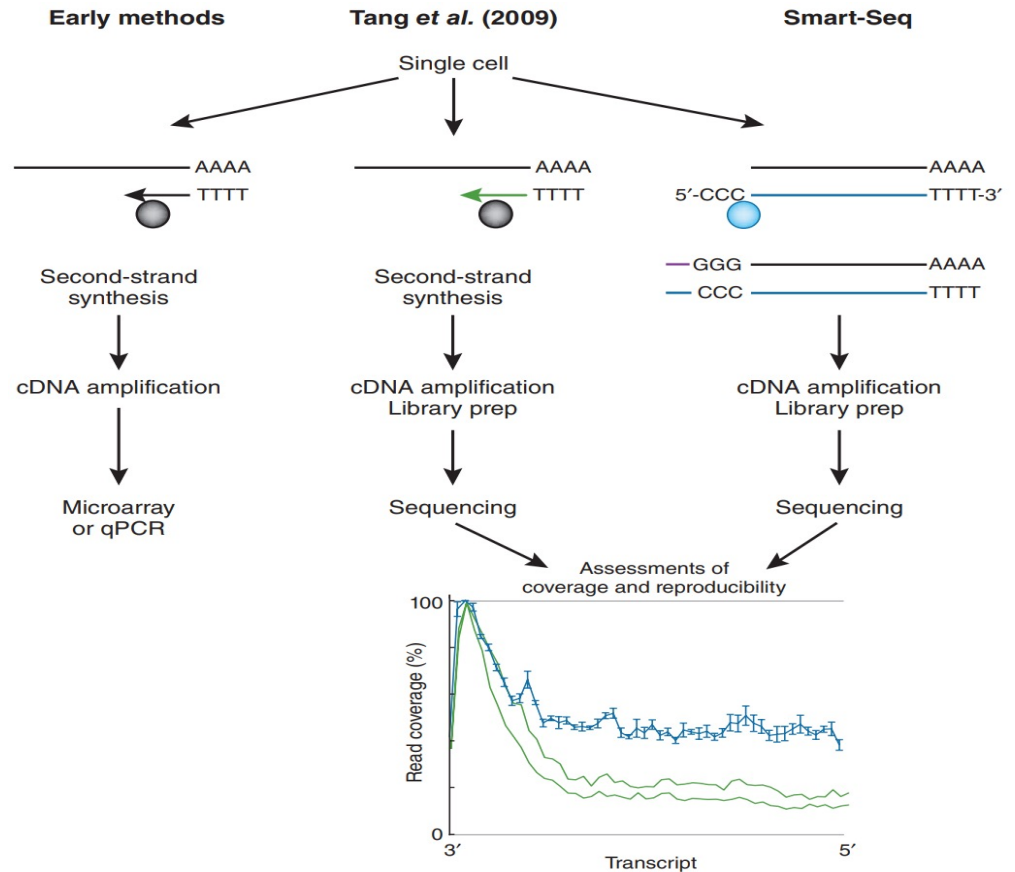
Published: 07 August 2012

Transcriptome sequencing of single cells with Smart-Seq

Jillian J Goetz & Jeffrey M Trimarchi

[Nature Biotechnology](#) **30**, 763–765 (2012) | [Cite this article](#)

10k Accesses | 55 Citations | 7 Altmetric | [Metrics](#)



2012年，单细胞全长转录组测序技术：SMART-seq

<https://www.nature.com/articles/nbt.2325>

STRT-seq

分子实验设计

nature methods

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[nature](#) > [nature methods](#) > [brief communications](#) > [article](#)

Published: 22 December 2013

Quantitative single-cell RNA-seq with unique molecular identifiers

[Saiful Islam](#), [Amit Zeisel](#), [Simon Joost](#), [Gioele La Manno](#), [Pawel Zajac](#), [Maria Kasper](#), [Peter Lönnerberg](#) & [Sten Linnarsson](#) 

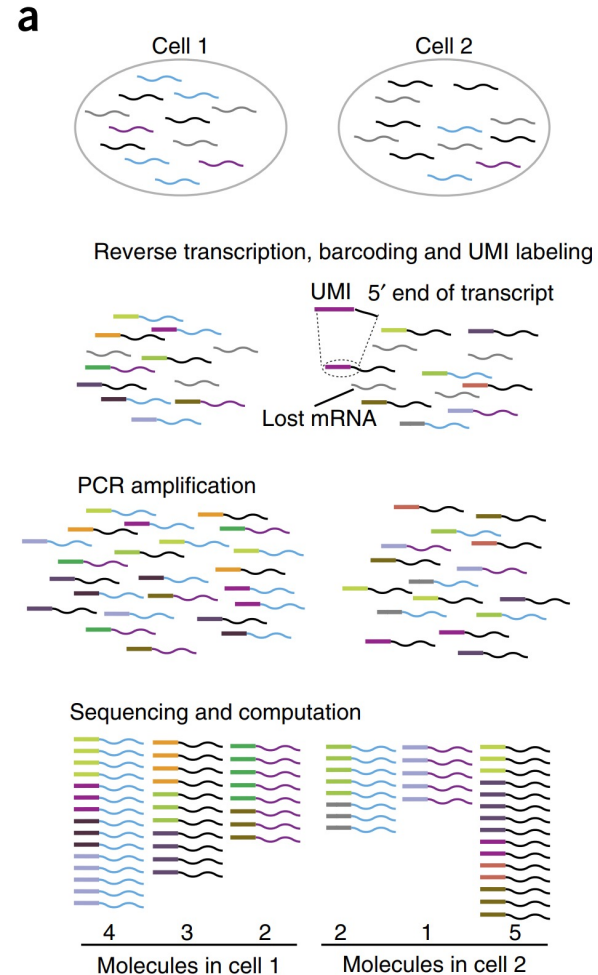
Nature Methods **11**, 163–166 (2014) | [Cite this article](#)

64k Accesses | 735 Citations | 48 Altmetric | [Metrics](#)



Sten Linnarsson
Karolinska Institute

2013年，首次引入UMI



<https://www.nature.com/articles/nmeth.2772>

SMART-seq2

分子实验设计

nature protocols

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[nature](#) > [nature protocols](#) > [protocols](#) > [article](#)

Published: 02 January 2014

Full-length RNA-seq from single cells using Smart-seq2

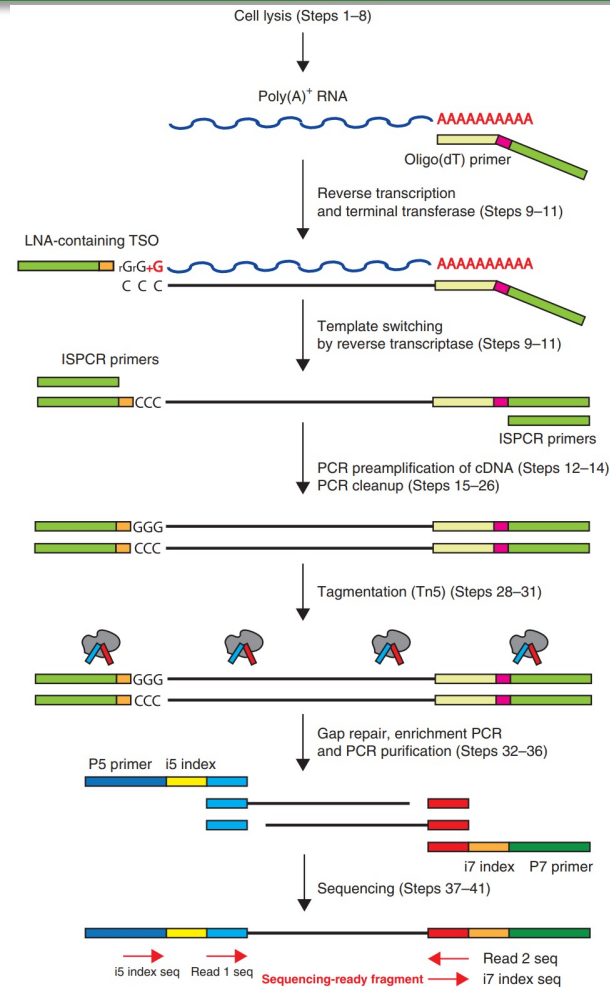
[Simone Picelli](#), [Omid R Faridani](#), [Åsa K Björklund](#), [Gösta Winberg](#), [Sven Sagasser](#) & [Rickard Sandberg](#) ✉

[Nature Protocols](#) **9**, 171–181 (2014) | [Cite this article](#)

143k Accesses | 1993 Citations | 75 Altmetric | [Metrics](#)



Rickard Sandberg
Karolinska Institute



2014年, SMART-seq2: TSO与模板结合热稳定性增强

<https://www.nature.com/articles/nprot.2014.006>

C1 microfluidics based method

细胞分离方式的改进



Stephen Quake

nature methods

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[nature](#) > [nature methods](#) > [analyses](#) > article

Published: 20 October 2013

Quantitative assessment of single-cell RNA-sequencing methods

[Angela R Wu](#), [Norma F Neff](#), [Tomer Kalisky](#), [Piero Dalerba](#), [Barbara Treutlein](#), [Michael E Rothenberg](#), [Francis M Mburu](#), [Gary L Mantalas](#), [Sopheak Sim](#), [Michael F Clarke](#) & [Stephen R Quake](#) ✉

Nature Methods **11**, 41–46 (2014) | [Cite this article](#)

61k Accesses | 505 Citations | 55 Altmetric | [Metrics](#)

C1 microfluidics based



Cells-to-Ct +
Fluidigm C1
STA kit
n = 184



Targeted
preamp



Multiplex
qPCR



SMARTer
n = 96



Nextera



RNA-seq

2014年，微流控分离细胞：C1

Drop-seq

细胞分离方式的改进

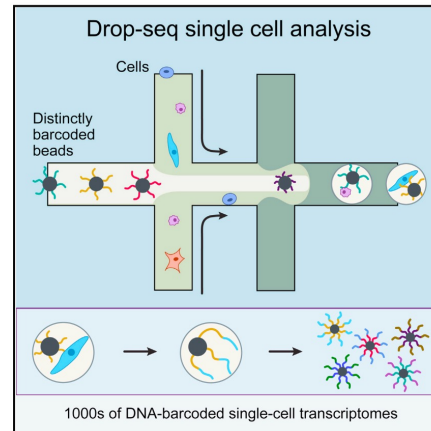


Steve McCarroll
Harvard University

Cell

Highly Parallel Genome-wide Expression Profiling of Individual Cells Using Nanoliter Droplets

Graphical Abstract



Authors

Evan Z. Macosko, Anindita Basu, ..., Aviv Regev, Steven A. McCarroll

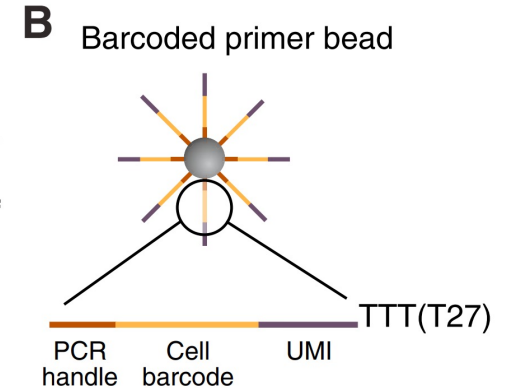
Correspondence

emacosko@genetics.med.harvard.edu (E.Z.M.),
mccarroll@genetics.med.harvard.edu (S.A.M.)

In Brief

Capturing single cells along with sets of uniquely barcoded primer beads together in tiny droplets enables large-scale, highly parallel single-cell transcriptomics. Applying this analysis to cells in mouse retinal tissue revealed transcriptionally distinct cell populations along with molecular markers of each type.

Resource



Synthesis of UMI (8 bases)

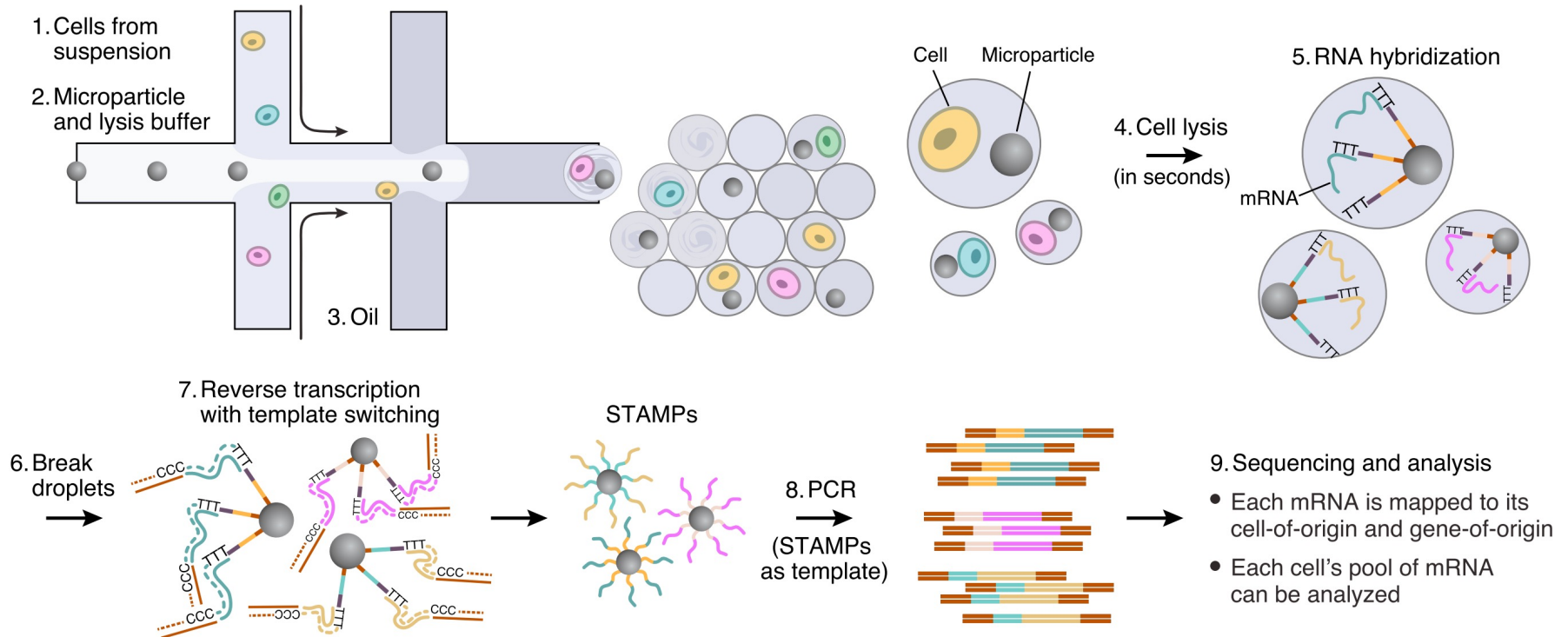


- Millions of the same **cell barcode** per bead
- 4^8 different **molecular barcodes** (UMIs) per bead

2015年，纳米液滴系统-油包水分离细胞：Drop-seq

Drop-seq

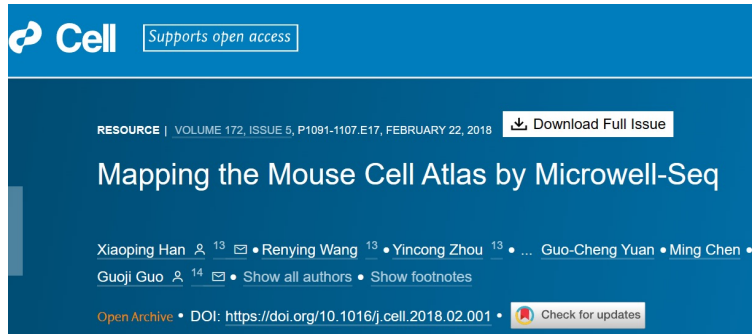
细胞分离方式的改进



Drop-seq

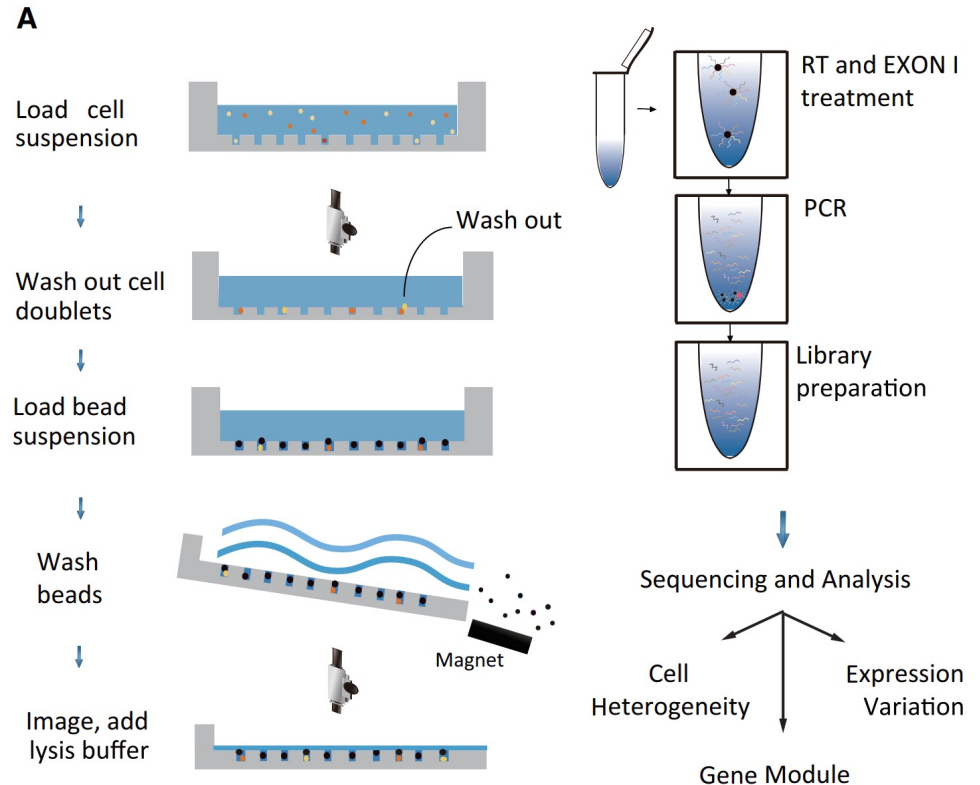
Microwell-seq

细胞分离方式的改进



郭国骥
浙江大学

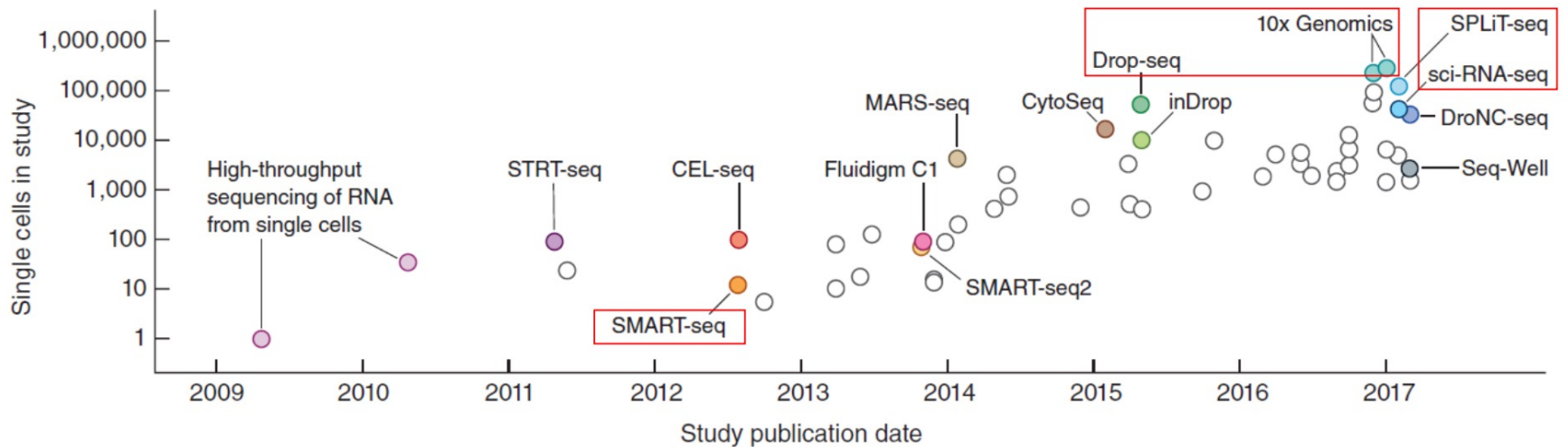
400, 000 single cells



2018年，微孔系统：Microwell-Seq

scRNA-seq技术发展时间轴

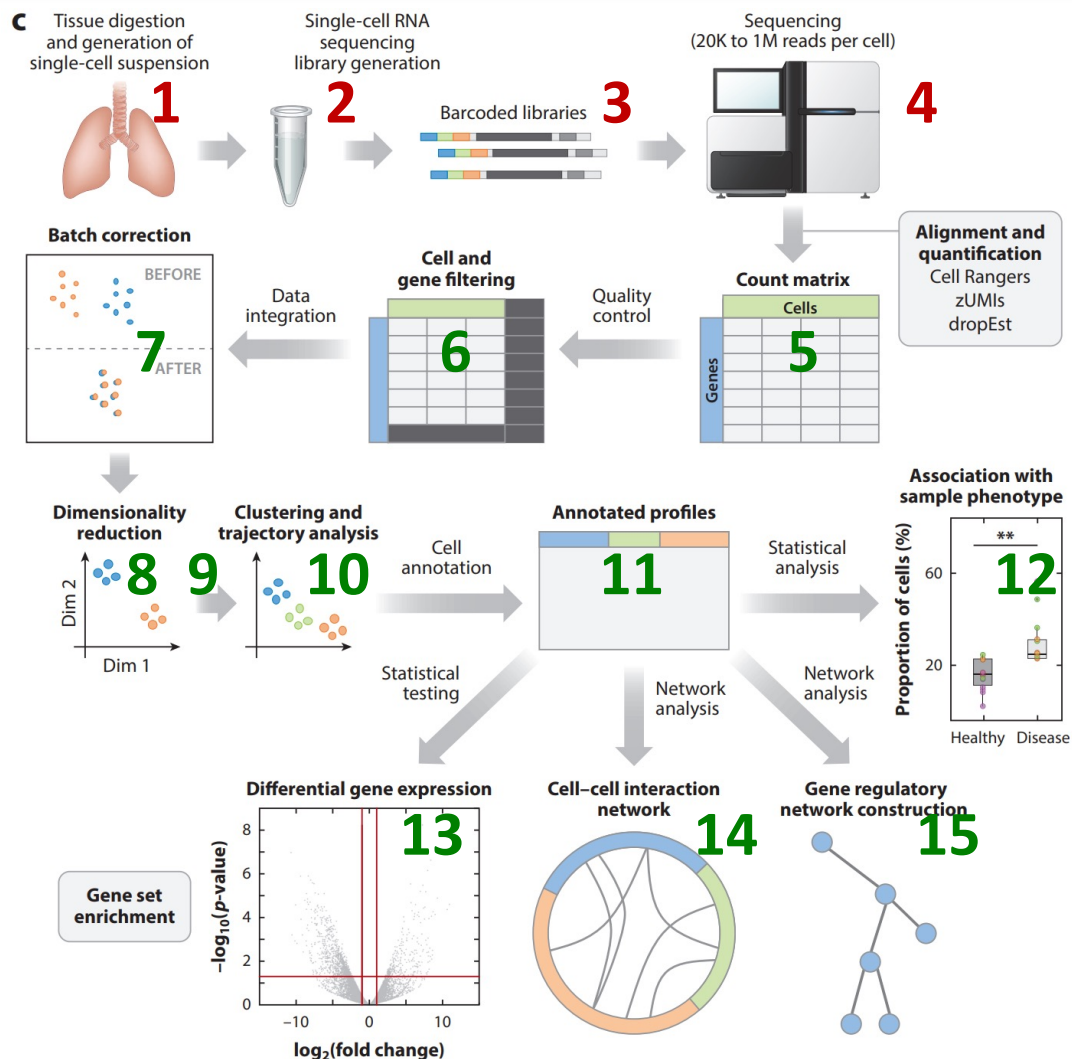
Cell numbers reported in representative publications by publication date. Key technologies are indicated.



Svensson V, Vento-Tormo R, Teichmann SA. Nat Protoc. 2018 Apr;13(4):599-604.

单细胞转录组技术及分析基本流程

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5. 比对-Cellranger



- 1.提取Cell barcode和UMI
- 2.数据质控:cutadapt
- 3.数据比对:STAR
- 4.转录本定量:featureCounts
- 5.获取单细胞基因表达数矩阵:umi_toolscount

5. 比对-Cellranger

Estimated Number of Cells

12,342

Mean Reads per Cell

28,019

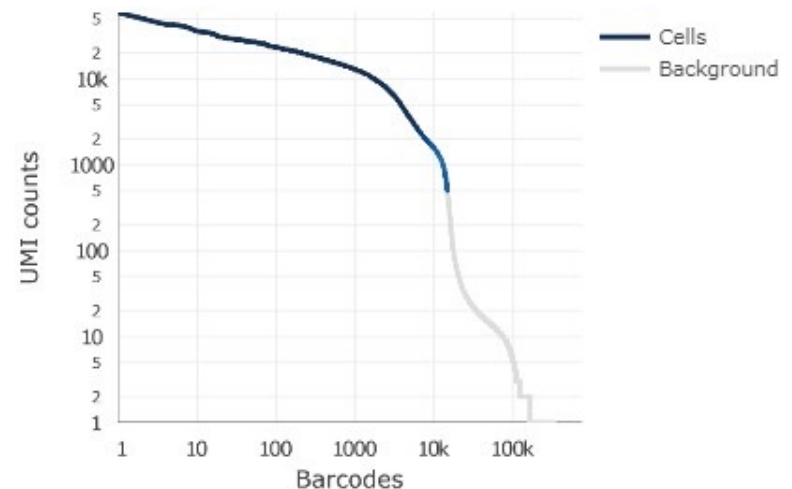
Median Genes per Cell

1,082

Sequencing

Number of Reads	345,812,670
Valid Barcodes	94.0%
Sequencing Saturation	73.2%
Q30 Bases in Barcode	96.6%
Q30 Bases in RNA Read	94.5%
Q30 Bases in RNA Read 2	93.1%
Q30 Bases in Sample Index	94.1%
Q30 Bases in UMI	96.3%

Cells



Estimated Number of Cells	12,342
Fraction Reads in Cells	92.3%
Mean Reads per Cell	28,019
Median Genes per Cell	1,082
Total Genes Detected	23,006
Median UMI Counts per Cell	2,594

Cellranger输入输出文件



■ 输入文件: [SampleName]_S1_L00[LaneNumber]_[ReadType]_001.fastq.gz

■ 输出文件:

- 比对文件
- 索引文件
- 矩阵文件

Name	
	barcodes.tsv
	genes.tsv
	matrix.mtx

6. 质控

QC and selecting cells for further analysis

■ 细胞的过滤

- 基因数:>200, 1k, 2k
- 线粒体比例:<5%

■ 基因的过滤

- 过滤不表达基因
- 细胞数:>3

■ 线粒体基因和红细胞基因比例的筛选

- 线粒体基因含量高的细胞可能是凋亡的细胞
 - 很少有细胞表达红细胞基因
-

Seurat - Guided Clustering Tutorial

■ https://satijalab.org/seurat/articles/pbmc3k_tutorial.html

Setup the Seurat Object

For this tutorial, we will be analyzing the a dataset of Peripheral Blood Mononuclear Cells (PBMC) freely available from 10X Genomics. There are 2,700 single cells that were sequenced on the Illumina NextSeq 500. The raw data can be found [here](#).

We start by reading in the data. The `Read10X()` function reads in the output of the `cellranger` pipeline from 10X, returning a unique molecular identified (UMI) count matrix. The values in this matrix represent the number of molecules for each feature (i.e. gene; row) that are detected in each cell (column). Note that more recent versions of `cellranger` now also output using the `h5 file format`, which can be read in using the `Read10X_h5()` function in Seurat.

We next use the count matrix to create a `Seurat` object. The object serves as a container that contains both data (like the count matrix) and analysis (like PCA, or clustering results) for a single-cell dataset. For more information, check out our [Seurat object interaction vignette], or our [GitHub Wiki](#). For example, in Seurat v5, the count matrix is stored in `pbmc[["RNA"]]$counts`.

```
library(dplyr)
library(Seurat)
library(patchwork)

# Load the PBMC dataset
pbmc.data <- Read10X(data.dir = "/brahms/mollag/practice/filtered_gene_bc_matrices/hg19/")
# Initialize the Seurat object with the raw (non-normalized data).
pbmc <- CreateSeuratObject(counts = pbmc.data, project = "pbmc3k", min.cells = 3, min.features = 200)
pbmc
```

Seurat - Guided Clustering Tutorial

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pbmc
```

Copy to clipboard



QC and selecting cells for further analysis

QC and selecting cells for further analysis

Seurat allows you to easily explore QC metrics and filter cells based on any user-defined criteria. A few QC metrics **commonly used** by the community include

- The number of unique genes detected in each cell.
 - Low-quality cells or empty droplets will often have very few genes
 - Cell doublets or multiplets may exhibit an aberrantly high gene count
- Similarly, the total number of molecules detected within a cell (correlates strongly with unique genes)
- The percentage of reads that map to the mitochondrial genome
 - Low-quality / dying cells often exhibit extensive mitochondrial contamination
 - We calculate mitochondrial QC metrics with the `PercentageFeatureSet()` function, which calculates the percentage of counts originating from a set of features
 - We use the set of all genes starting with `MT-` as a set of mitochondrial genes

```
# The [[ operator can add columns to object metadata. This is a great place to stash QC stats
pbmc[["percent.mt"]] <- PercentageFeatureSet(pbmc, pattern = "^MT-")
```

► **Where are QC metrics stored in Seurat?**

QC and selecting cells for further analysis

```
# The [[ operator can add columns to object metadata. This is a great place to stash QC stats
pbmc[["percent.mt"]] <- PercentageFeatureSet(pbmc, pattern = "^MT-")
```

▼ Where are QC metrics stored in Seurat?

- The number of unique genes and total molecules are automatically calculated during `CreateSeuratObject()`
 - You can find them stored in the object meta data

```
# Show QC metrics for the first 5 cells
head(pbmc@meta.data, 5)
```

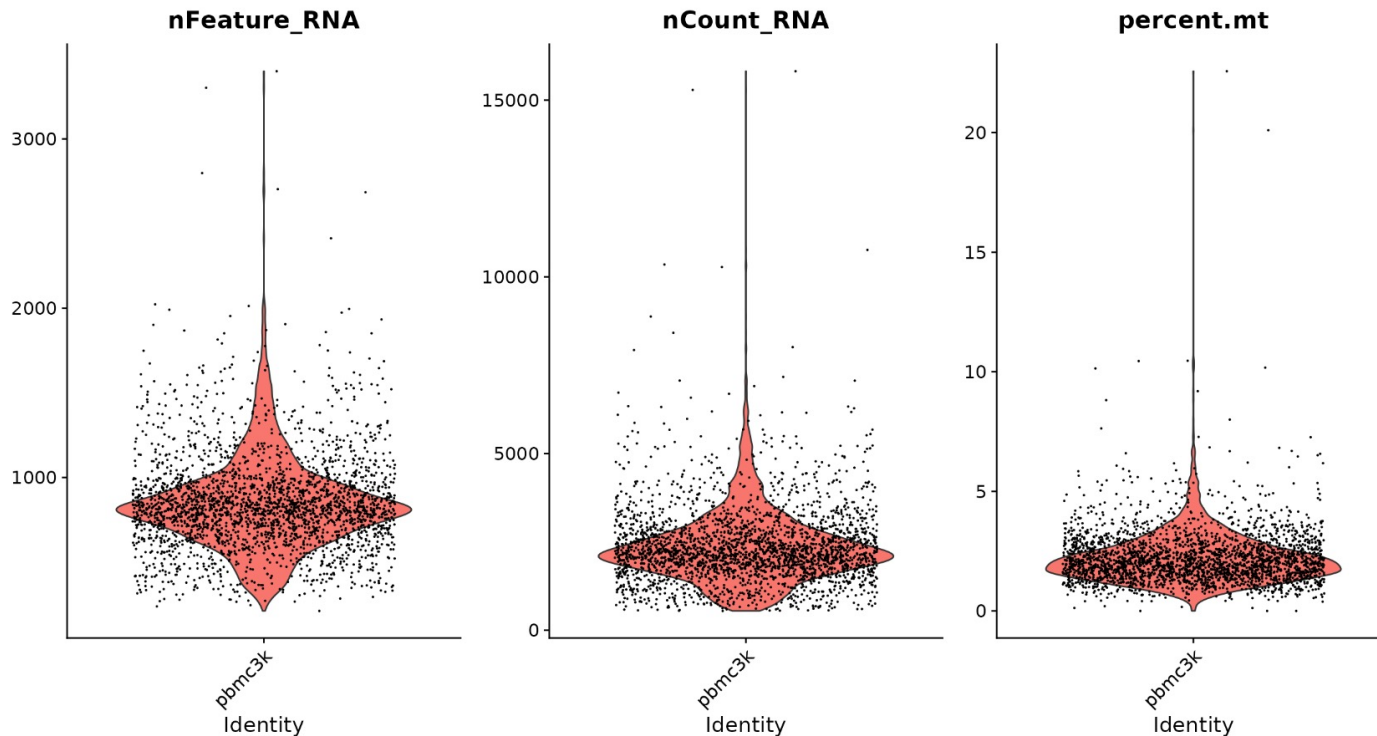
##	orig.ident	nCount_RNA	nFeature_RNA	percent.mt
## AAACATACAACCAC-1	pbmc3k	2419	779	3.0177759
## AAACATTGAGCTAC-1	pbmc3k	4903	1352	3.7935958
## AAACATTGATCAGC-1	pbmc3k	3147	1129	0.8897363
## AAACCGTGCTTCCG-1	pbmc3k	2639	960	1.7430845
## AAACCGTGTATGCG-1	pbmc3k	980	521	1.2244898

QC and selecting cells for further analysis

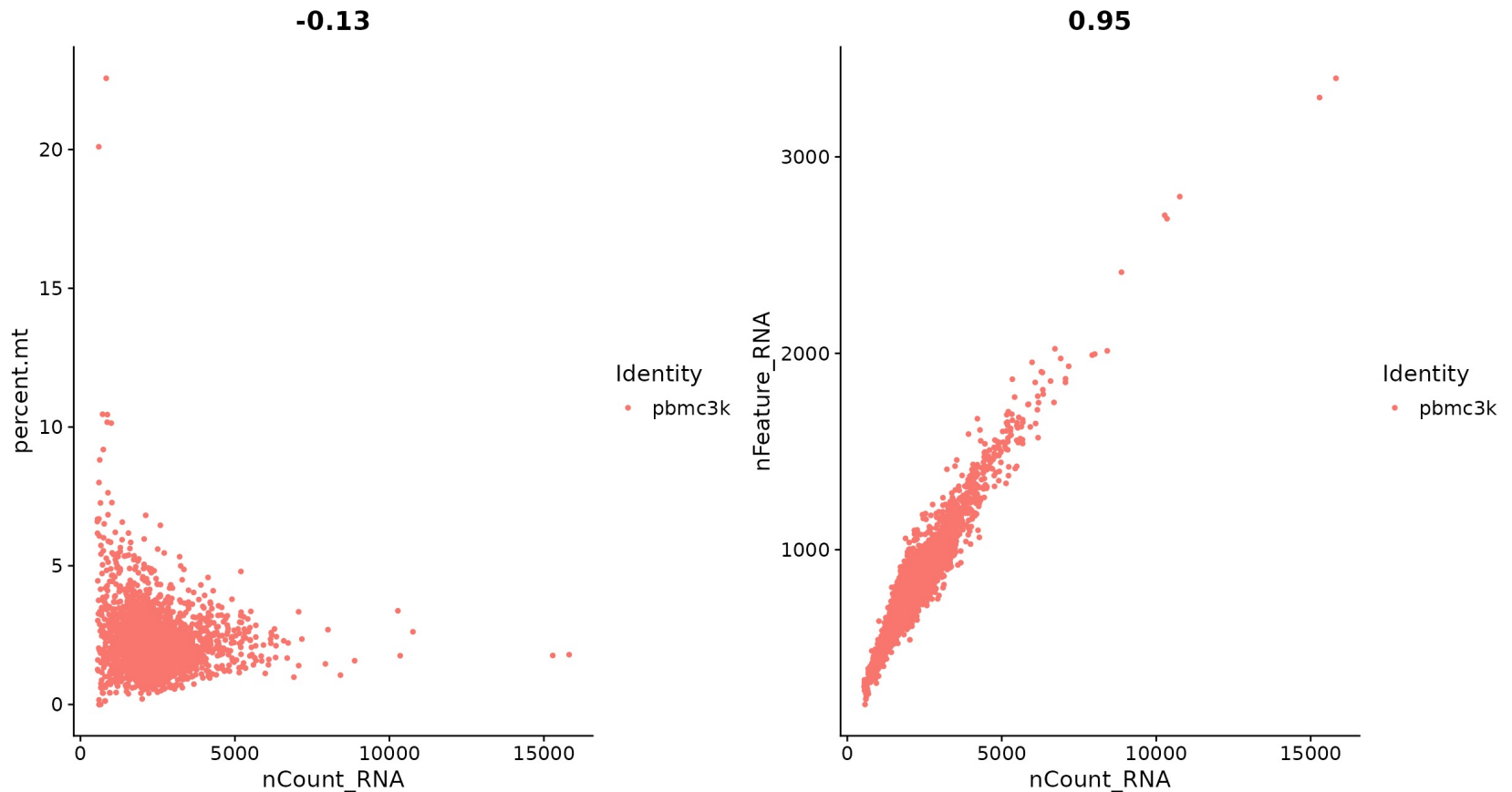
In the example below, we visualize QC metrics, and use these to filter cells.

- We filter cells that have unique feature counts over 2,500 or less than 200
- We filter cells that have >5% mitochondrial counts

```
# Visualize QC metrics as a violin plot  
VlnPlot(pbmc, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)
```



FeatureScatter is typically used to visualize feature-feature relationships



Normalizing the data (单细胞数据的标准化)

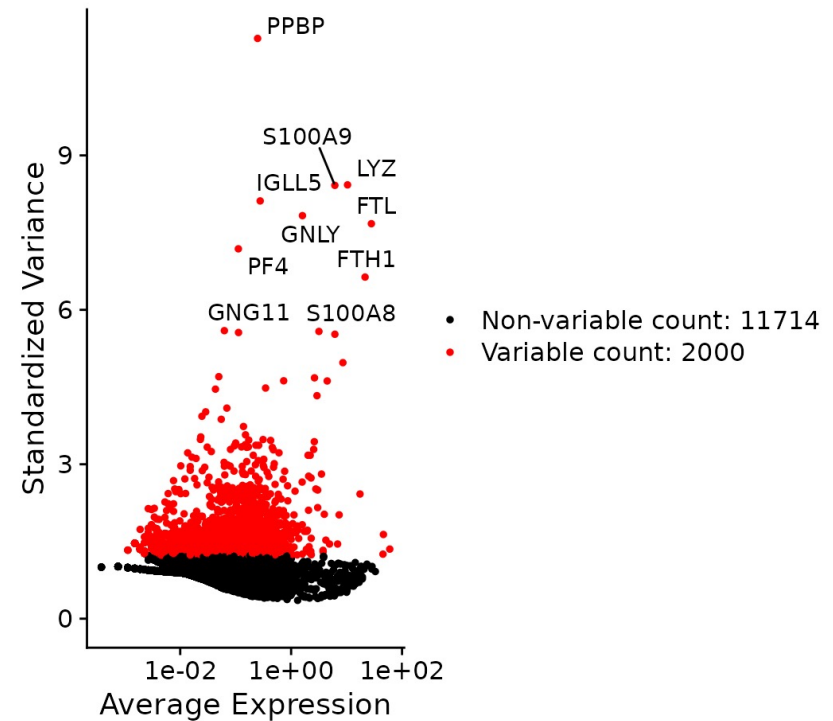
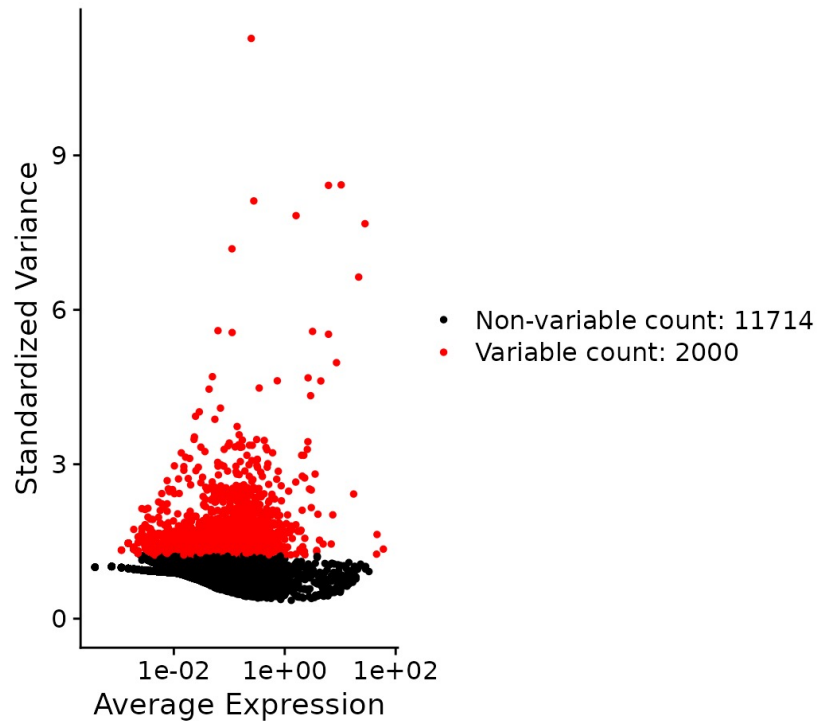
■ 常见标准化方法——CPM

- 从取样到建库上机每个操作步骤的可变性，即便同一个细胞测序两次获得的计数深度也可能会有所不同。因此，当基于原始计数数据比较细胞之间的基因表达时，得到的差异可能来自于技术原因。
- 使得细胞之间可以相互比较
- 测序深度 (sequencing depth) 标准化
 - Downsampling: 抽取等量reads
 - **CPM (count per million)**
- $CPM = 10^6 * C_{ij} / \sum_j C_{ij}$
- $data = \log((CPM/10) + 1)$

$$CPM = \frac{\text{Number of reads mapped to gene} \times 10^6}{\text{Total number of mapped reads}}$$

Seurat的默认**标准化方法**是:每个细胞的某一count先除以该细胞的总count，然后乘以scale因子-10000，再做个对数转换。

Identification of highly variable features (feature selection)



单细胞数据的归一化(scale/z-score)

- **基因归一化**是指一个基因减去其在所有样品表达的均值然后除以其在所有样品表达值的标准差。

$$x'_i = \frac{x_i - \bar{x}}{s}$$

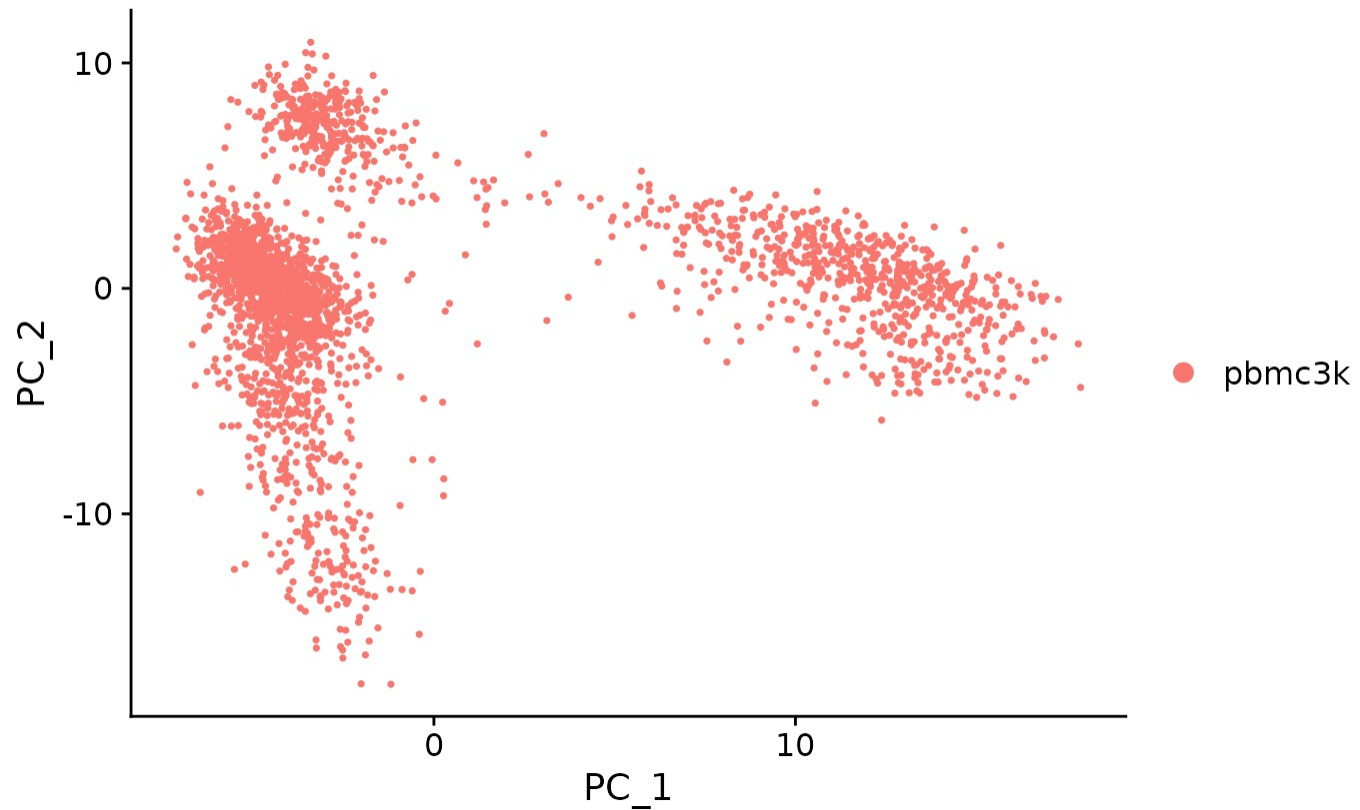
Scaling the data

Next, we apply a linear transformation ('scaling') that is a standard pre-processing step prior to dimensional reduction techniques like PCA. The `ScaleData()` function:

- Shifts the expression of each gene, so that the mean expression across cells is 0
- Scales the expression of each gene, so that the variance across cells is 1
 - This step gives equal weight in downstream analyses, so that highly-expressed genes do not dominate
- The results of this are stored in `pbmc[["RNA"]].@scale.data`

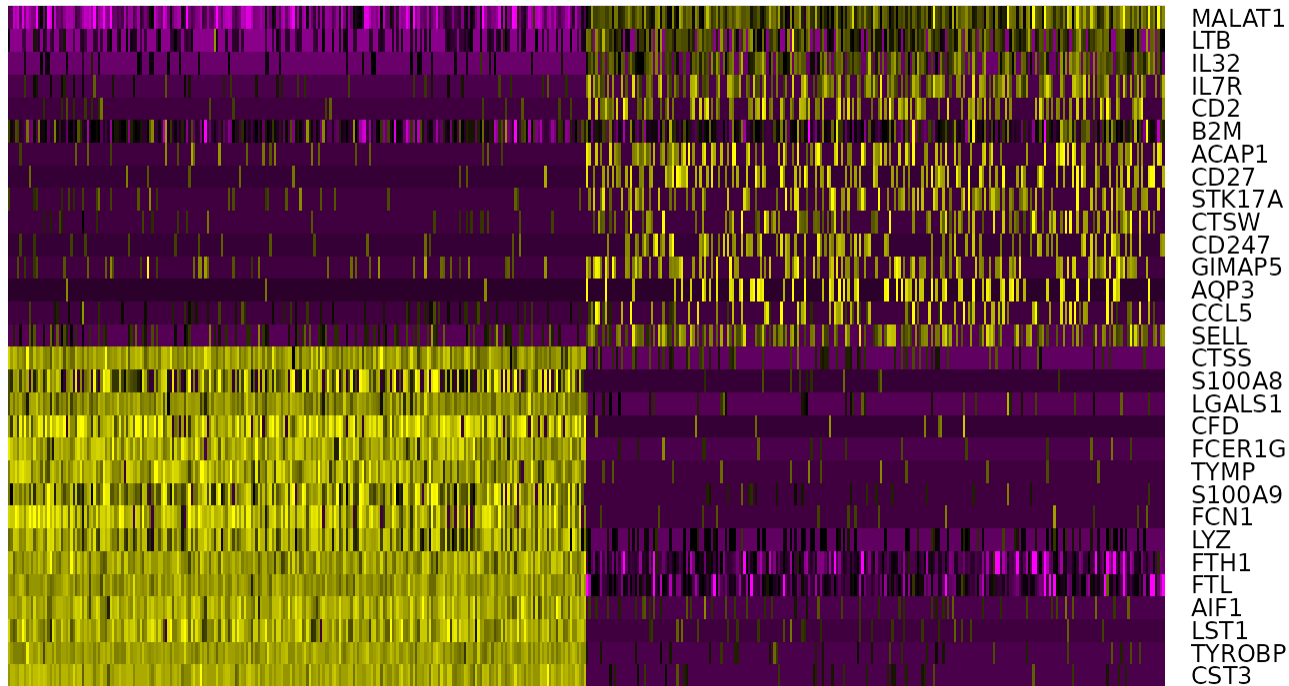
```
all.genes <- rownames(pbmc)
pbmc <- ScaleData(pbmc, features = all.genes)
```


Perform linear dimensional reduction



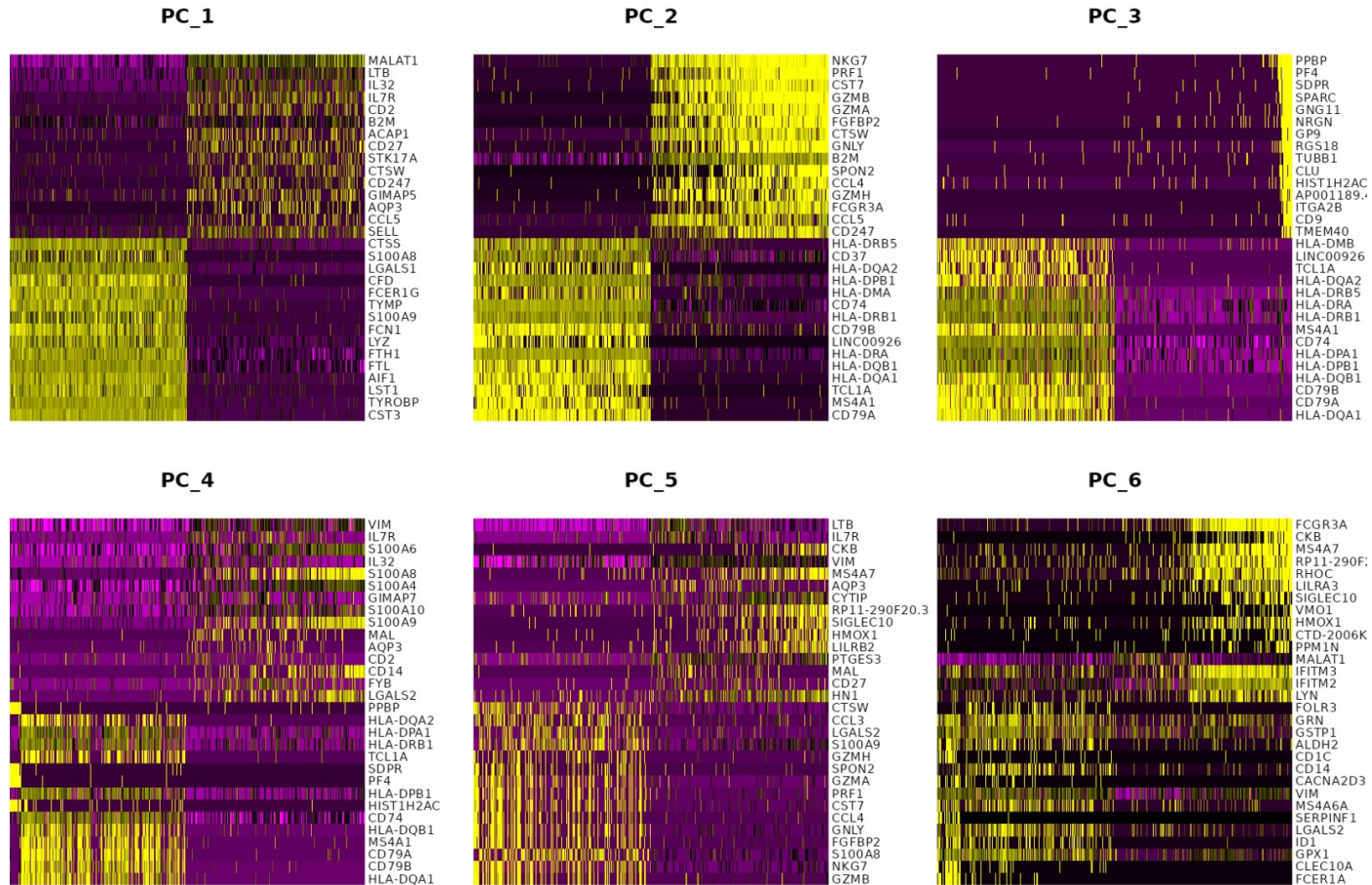
Perform linear dimensional reduction

PC_1



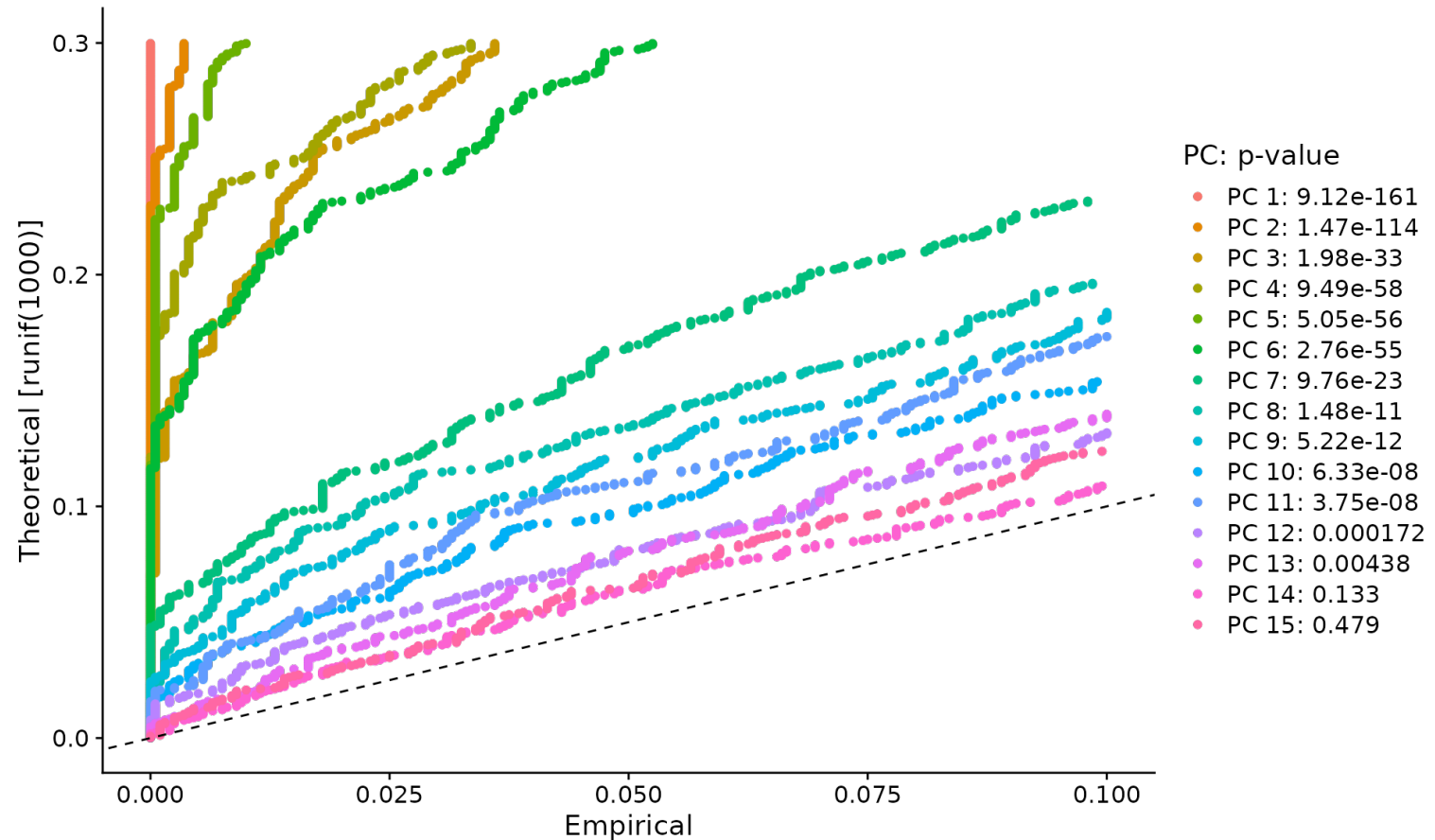
```
DimHeatmap(pbmcc, dims = 1, cells = 500, balanced = TRUE)
```

Perform linear dimensional reduction



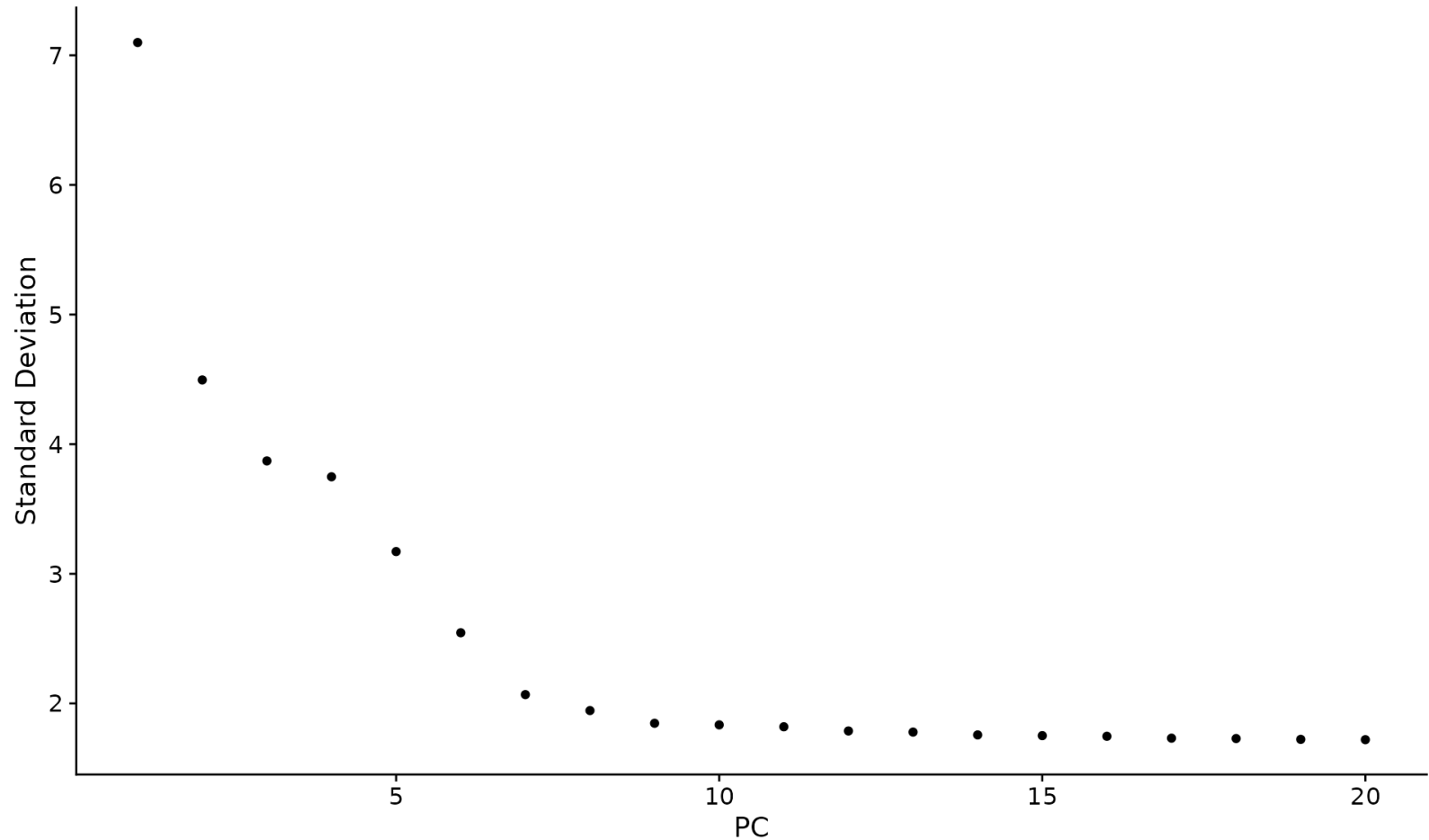
```
DimHeatmap(pbmcc, dims = 1, cells = 500, balanced = TRUE)
```

Determine the 'dimensionality' of the dataset



```
JackStrawPlot(pbmc, dims = 1:15)
```

Determine the 'dimensionality' of the dataset



Cluster the cells

- Briefly, these methods embed cells in a graph using a K-nearest neighbor (KNN) graph, with edges connecting cells with similar feature expression patterns, and then partition this graph into highly interconnected 'communities'.
- As in PhenoGraph, we first construct a KNN graph using euclidean distance in PCA space, and refine it based on shared neighborhoods (Jaccard similarity). This is done by the **FindNeighbors()** function, and takes the defined dimensionality of the dataset (first 10 principal components) as input.

单细胞基因表达矩阵（高维稀疏）



▼ PCA降维（保留主要变异）

低维PCA空间中的细胞点云



▼ 计算细胞间欧氏距离 + 找K近邻

KNN图（细胞为节点，相似性为边）



▼ Jaccard相似度精炼边权重

SNN图（更鲁棒的邻域结构）



▼ Louvain/Leiden社区检测

细胞簇（clusters）≈ 细胞类型

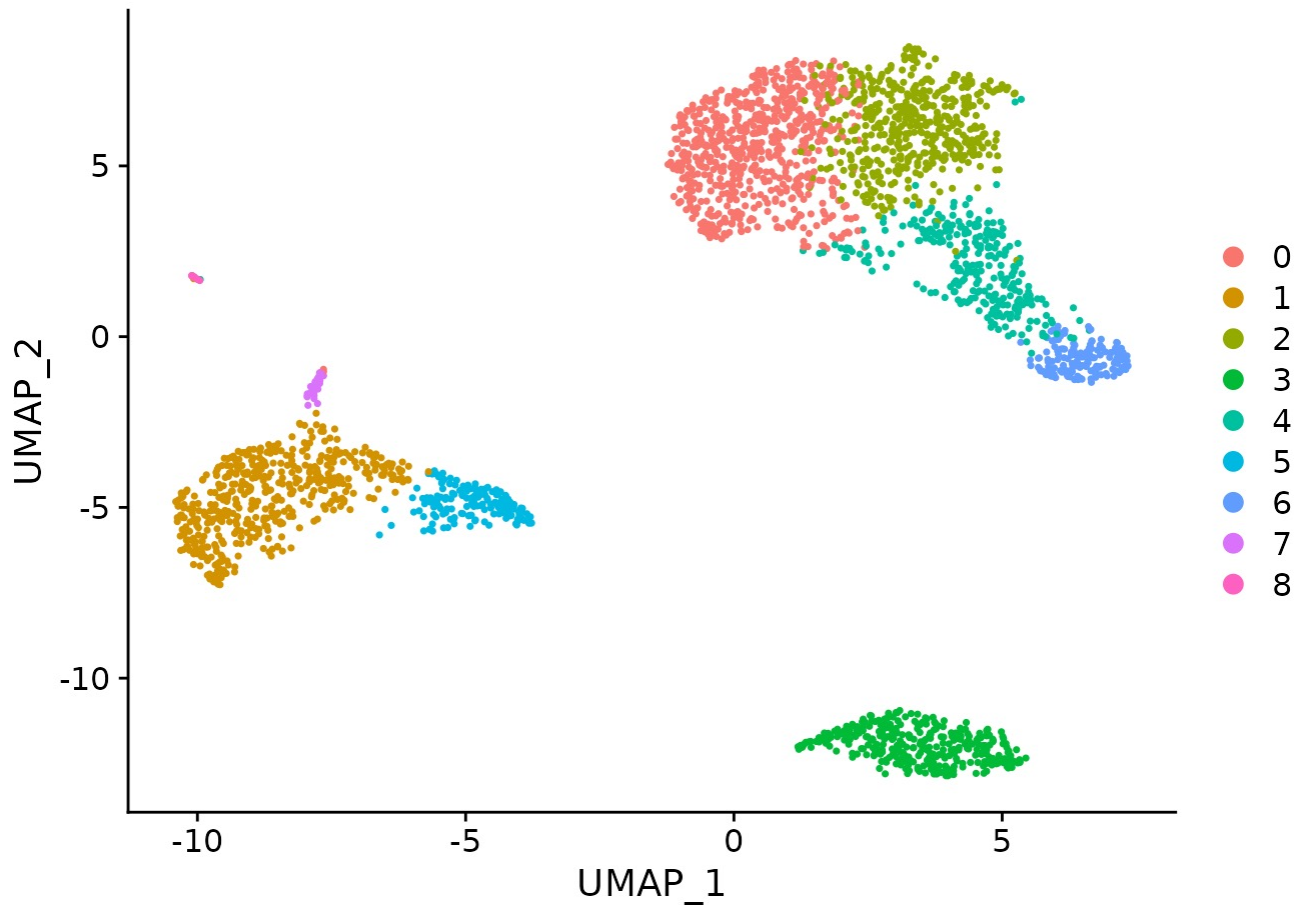


▼ UMAP可视化

直观的二维细胞图谱

```
pbmc <- FindNeighbors(pbmc, dims = 1:10)
pbmc <- FindClusters(pbmc, resolution = 0.5)
```

Run non-linear dimensional reduction (UMAP/tSNE)



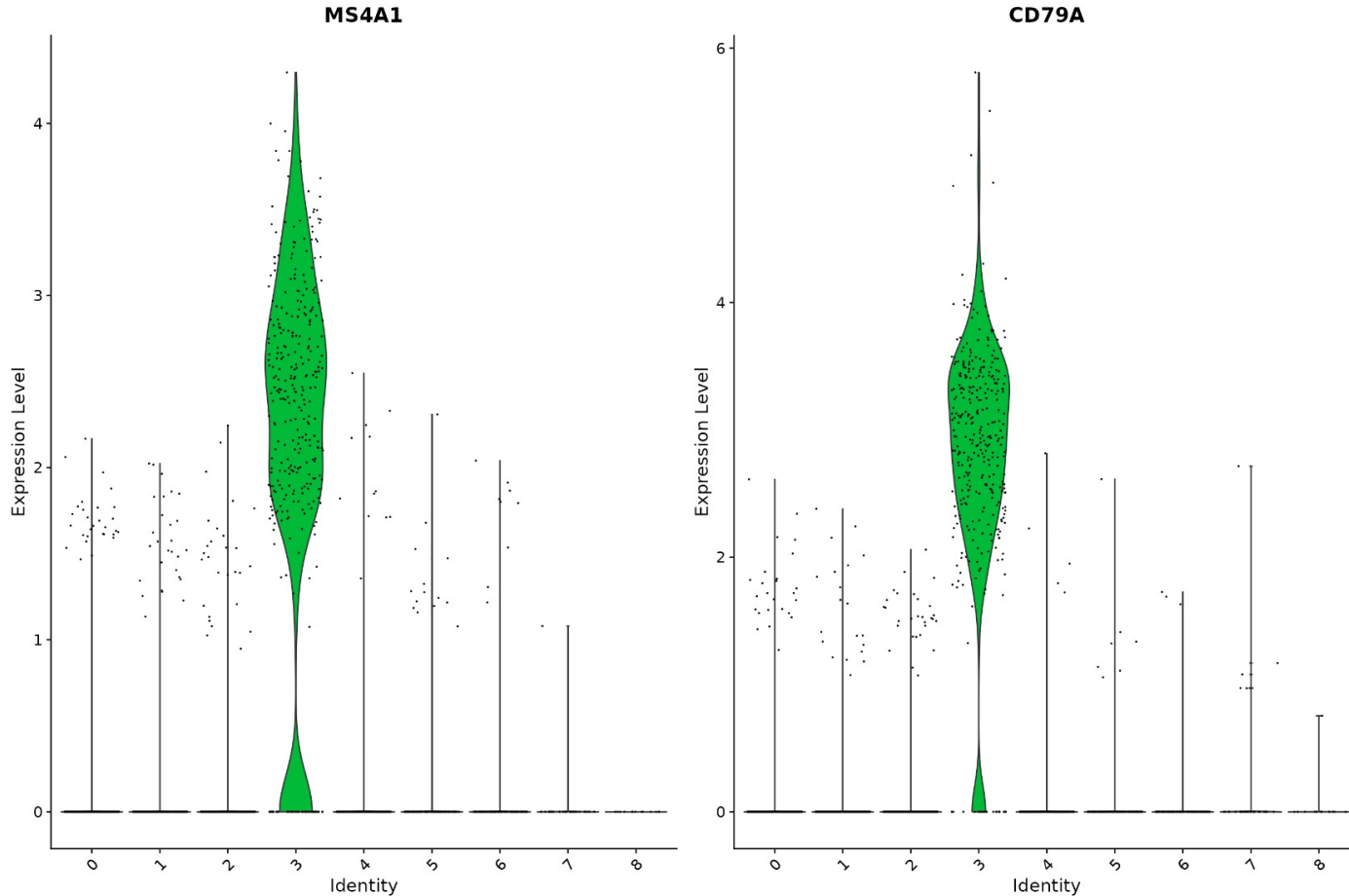
Finding differentially expressed features (cluster biomarkers)

```
# find all markers of cluster 2
cluster2.markers <- FindMarkers(pbmc, ident.1 = 2, min.pct = 0.25)
head(cluster2.markers, n = 5)
```

```
##           p_val avg_log2FC pct.1 pct.2    p_val_adj
## IL32 2.892340e-90  1.2013522 0.947 0.465 3.966555e-86
## LTB  1.060121e-86  1.2695776 0.981 0.643 1.453850e-82
## CD3D 8.794641e-71  0.9389621 0.922 0.432 1.206097e-66
## IL7R 3.516098e-68  1.1873213 0.750 0.326 4.821977e-64
## LDHB 1.642480e-67  0.8969774 0.954 0.614 2.252497e-63
```

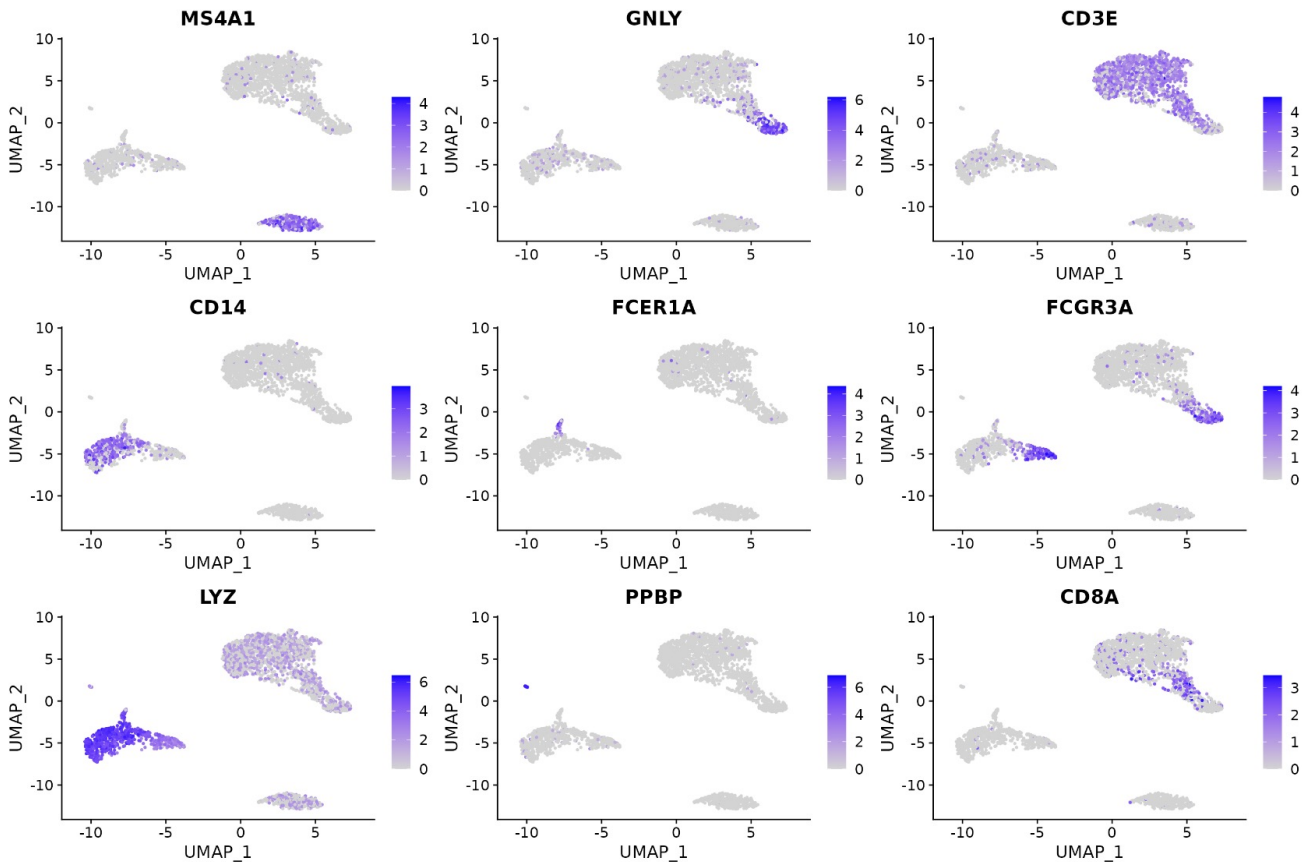
visualizing marker expression

```
VlnPlot(pbm, features = c("MS4A1", "CD79A"))
```



visualizing marker expression

```
FeaturePlot(pbm, features = c("MS4A1", "GNLY", "CD3E", "CD14", "FCER1A", "FCGR3A", "LYZ", "PPBP",  
"CD8A"))
```

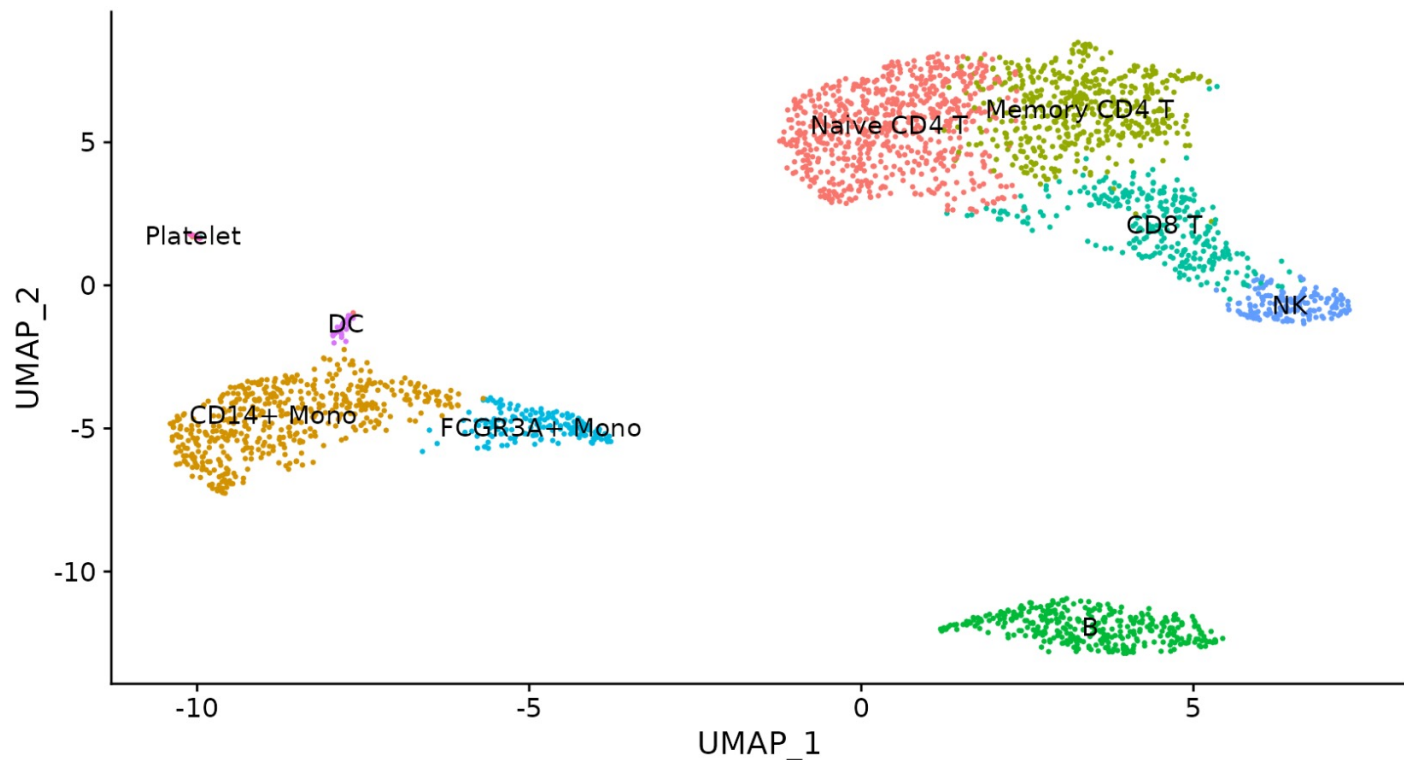


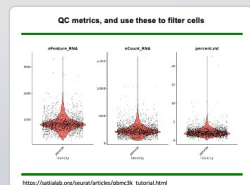
Assigning cell type identity to clusters

Cluster ID	Markers	Cell Type
0	IL7R, CCR7	Naive CD4+ T
1	CD14, LYZ	CD14+ Mono
2	IL7R, S100A4	Memory CD4+
3	MS4A1	B
4	CD8A	CD8+ T
5	FCGR3A, MS4A7	FCGR3A+ Mono
6	GNLY, NKG7	NK
7	FCER1A, CST3	DC
8	PPBP	Platelet

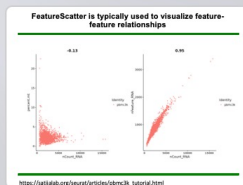
Assigning cell type identity to clusters

```
new.cluster.ids <- c("Naive CD4 T", "CD14+ Mono", "Memory CD4 T", "B", "CD8 T", "FCGR3A+ Mono",  
  "NK", "DC", "Platelet")  
names(new.cluster.ids) <- levels(pbmc)  
pbmc <- RenameIdents(pbmc, new.cluster.ids)  
DimPlot(pbmc, reduction = "umap", label = TRUE, pt.size = 0.5) + NoLegend()
```

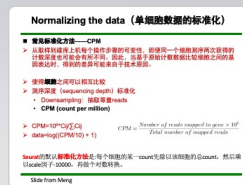




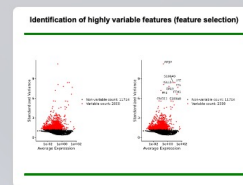
19



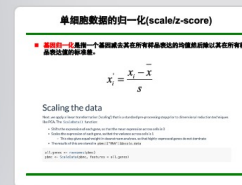
20



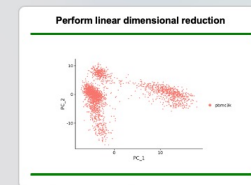
21



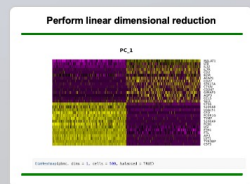
22



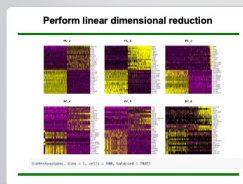
23



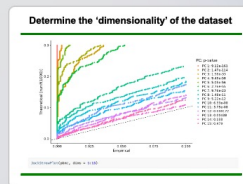
24



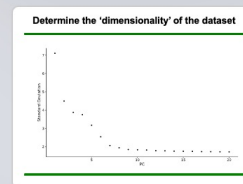
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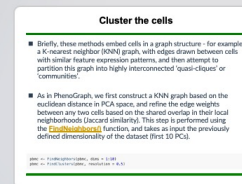
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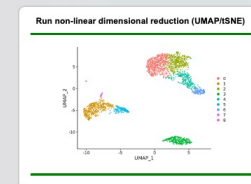
27



28



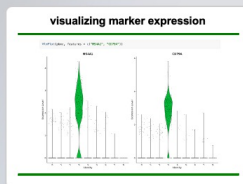
29



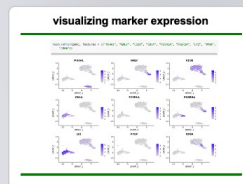
30



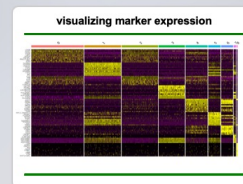
31



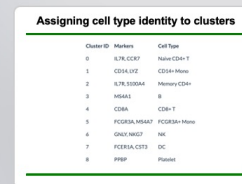
32



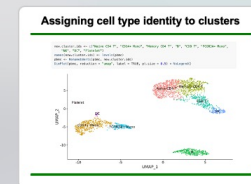
33



34



35



36

单细胞数据的降维

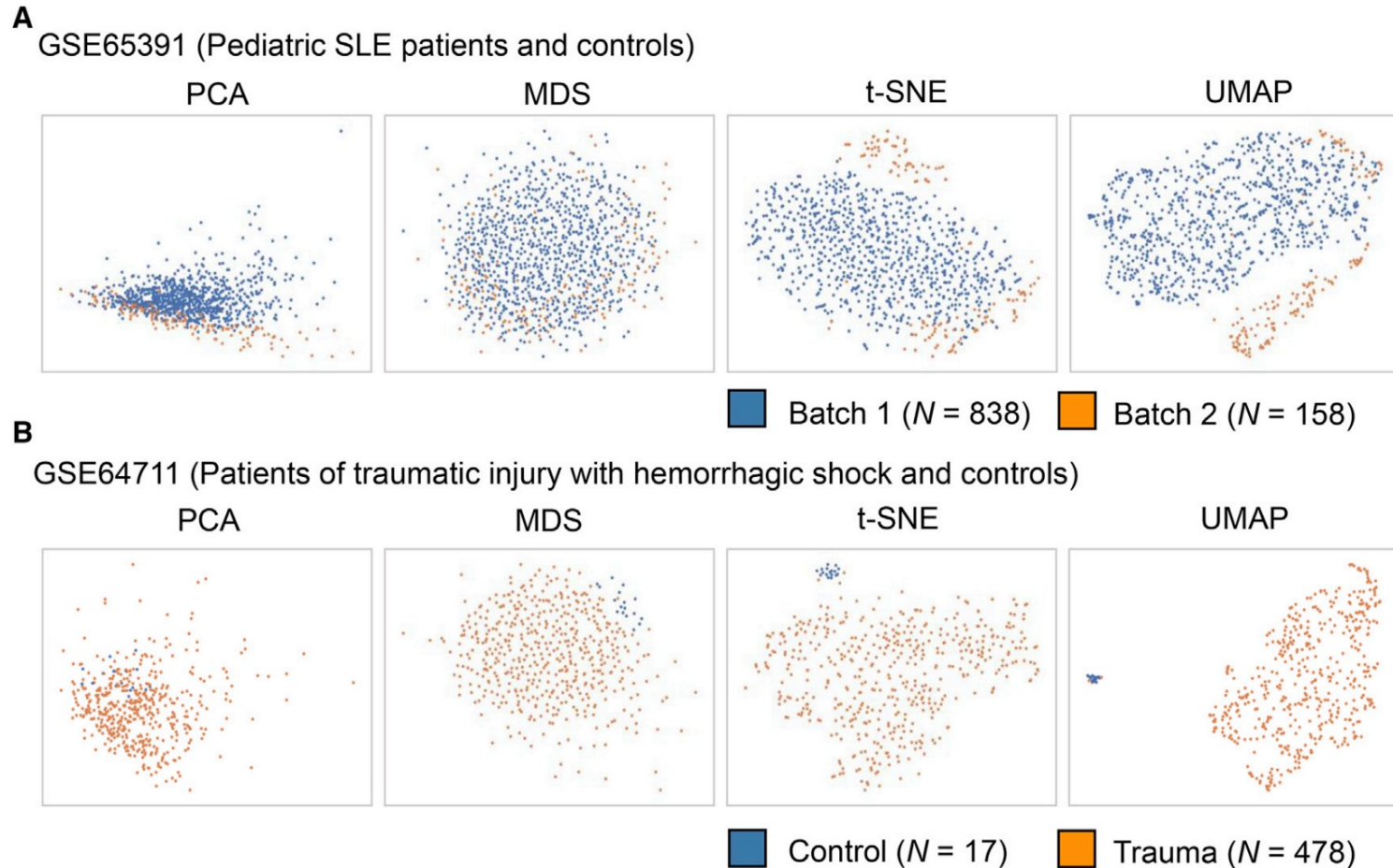


Figure 3. Biological explanation of clustering by batch effects and biological group

手动注释1： 利用差异基因注释

■ 差异基因标记细胞类型

- • 转录因子:影响谱系lineage分化，遗传操作可控制该细胞类型
- • 表面蛋白:抗体富集该种细胞类型
- • 特异表达基因或基因集组合

■ 如何找标志基因

- 看文献收集:review文章;
- 相关领域单细胞文章。
- • 数据库
- • 实验室传承

■ 红细胞:

- `c('HBG1','HBG2','HBQ1','HBA1','HBA2','HBE1','HBD','HBM','HBZ','HBB')`
-

手动注释1： 利用差异基因注释-数据库

- CellMarker <http://xteam.xbio.top/CellMarker/>



[Home](#)

Welcome to CellMarker

- Cell Taxonomy
<https://ngdc.cncb.ac.cn/celltaxonomy/>

 CNCB-NGDC

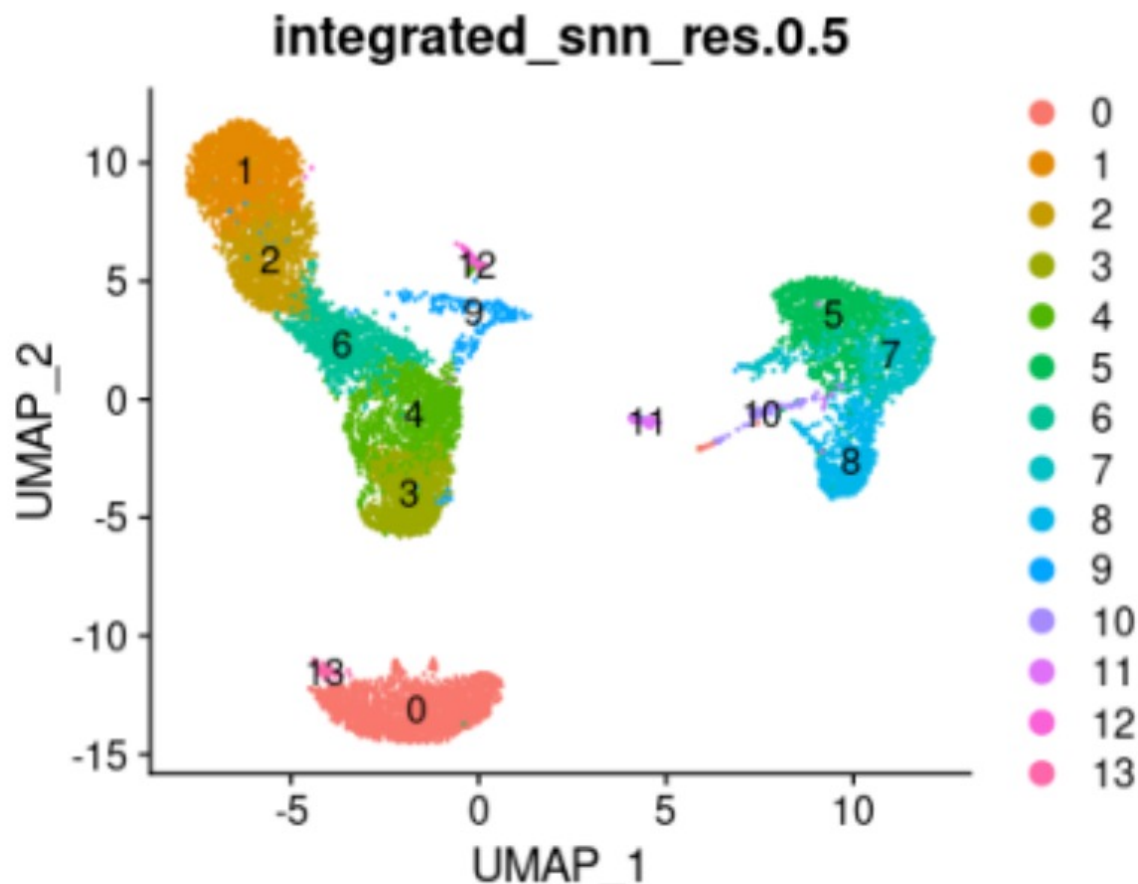


Cell Taxonomy

[Home](#)

手动注释1： 利用差异基因注释

- 比对网站中的基因
- 手动标记每个细胞簇对应的细胞类型

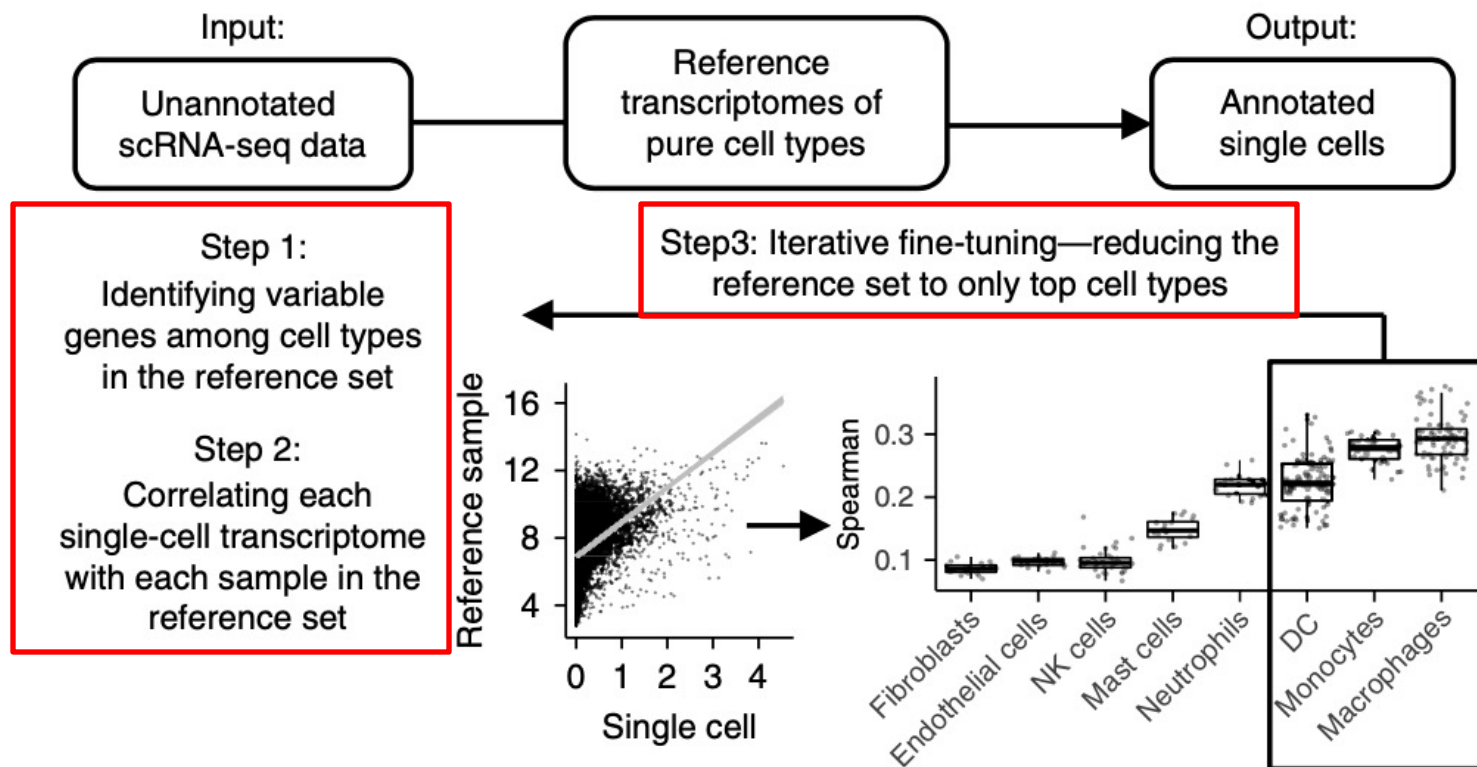


The bubble plot displays the relationship between 15 features and 14 identities. The x-axis represents features, and the y-axis represents identities. The size of each bubble indicates the 'Percent Expressed' (0, 25, 50, 75), and the color indicates the 'Average Expression' (0 to 2). The plot shows that certain features are highly expressed in specific identities, such as NKG7 in identity 13 and CD14 in identity 7.

自动注释-SingleR

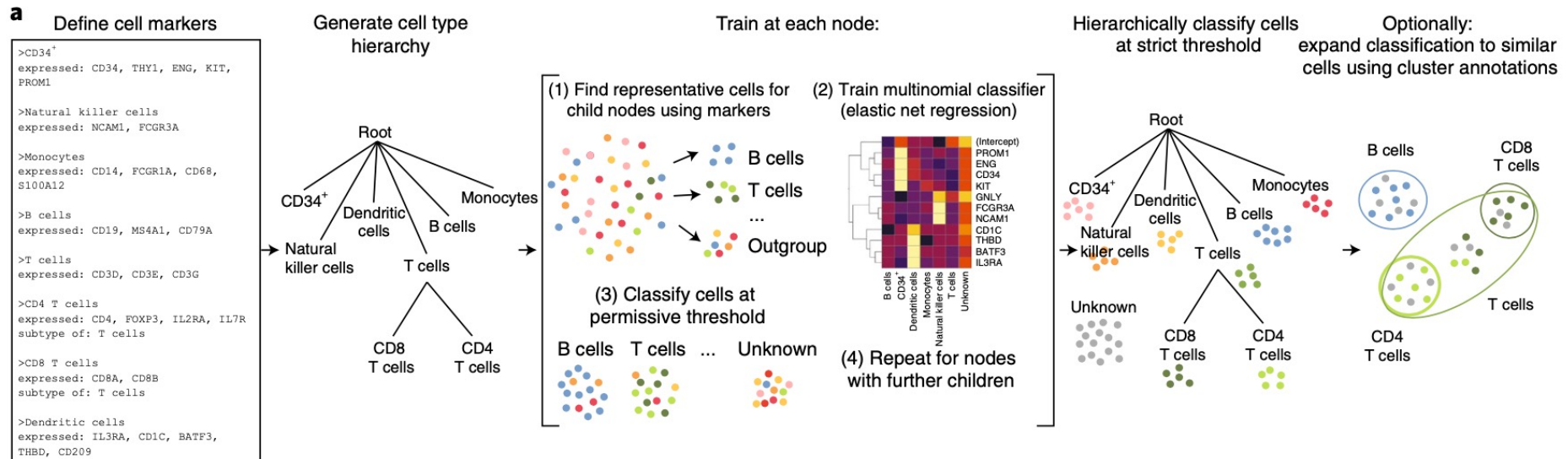
■ 对于每个测试单元：

1. 计算其表达谱与每个参考样品的表达谱之间的Spearman相关性。
2. 将每个标签的分数定义为相关性分布的fixed quantile（默认为0.8）。
3. 对所有标签重复此操作，然后将得分最高的标签作为此细胞的注释。



自动注释-Garnett

基于细胞类型特异性基因的可解释的分层标记语言，用于在单细胞转录分析和单细胞染色质可及性数据集中快速注释细胞类型



构建细胞类型标记文件

生成细胞类型树

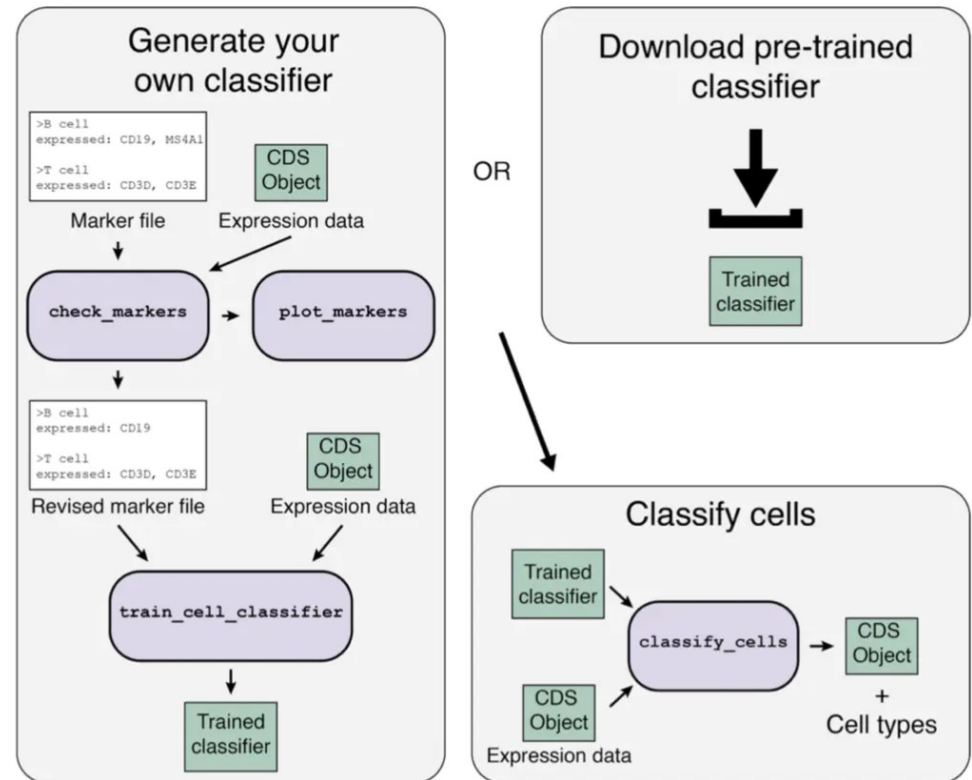
训练分类器

完成细胞分类

自动注释-Garnett

1. 制作符合garnett格式要求的定义细胞类型的marker基因文本文件(marker file)。
2. 使用单细胞数据创建monocle3的CDS数据对象(cds object)。
3. 将marker file和cds object输入garnett, 对marker基因进行打分, 然后根据评分结果优化marker file。
4. 将优化后的marker file和cds object输入garnett训练分类器。使用分类器对新的cds数据进行细胞分类。

Garnett overview



拟时序分析

[Published: 23 March 2014](#)

The dynamics and regulators of cell fate decisions are revealed by pseudotemporal ordering of single cells

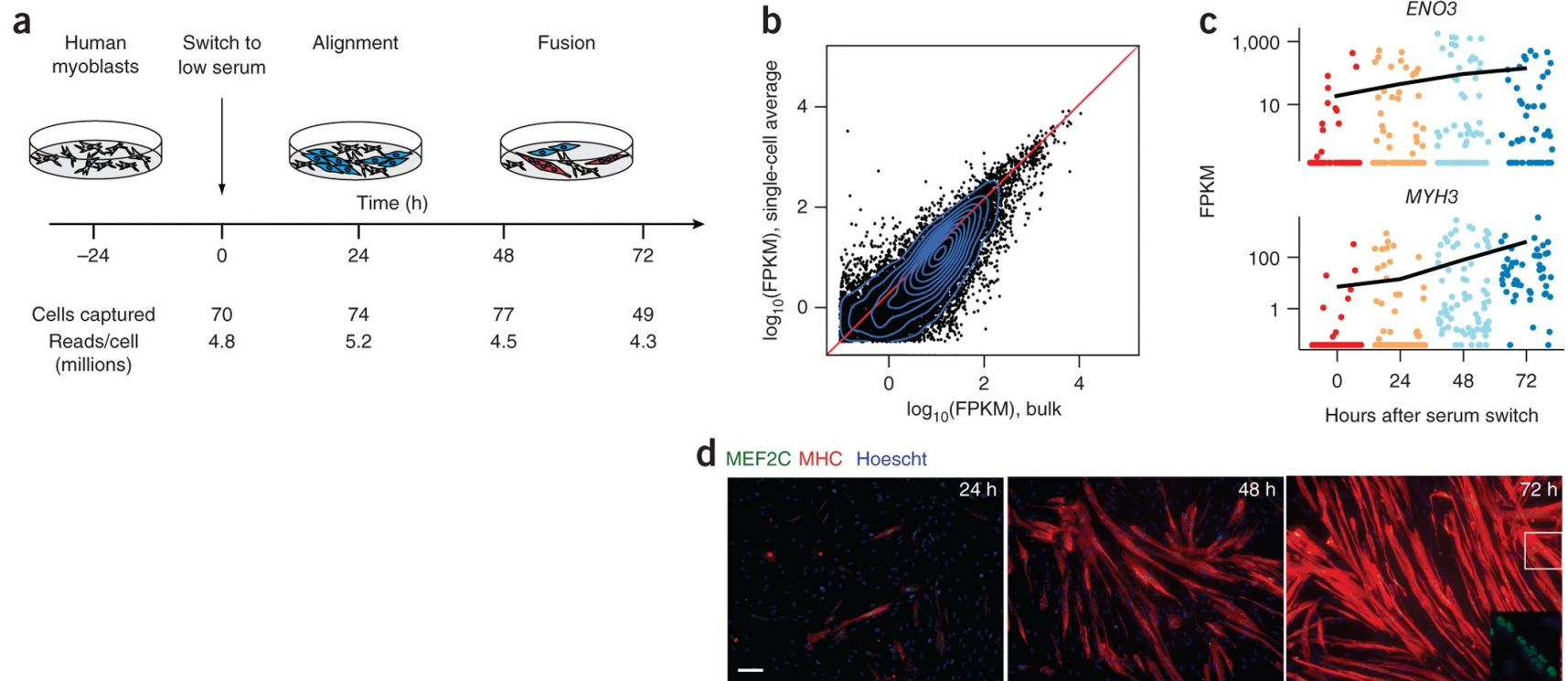
[Cole Trapnell](#), [Davide Cacchiarelli](#), [Jonna Grimsby](#), [Prapti Pokharel](#), [Shuqiang Li](#), [Michael Morse](#), [Niall J Lennon](#), [Kenneth J Livak](#), [Tarjei S Mikkelsen](#) & [John L Rinn](#) 

[Nature Biotechnology](#) **32**, 381–386 (2014) | [Cite this article](#)

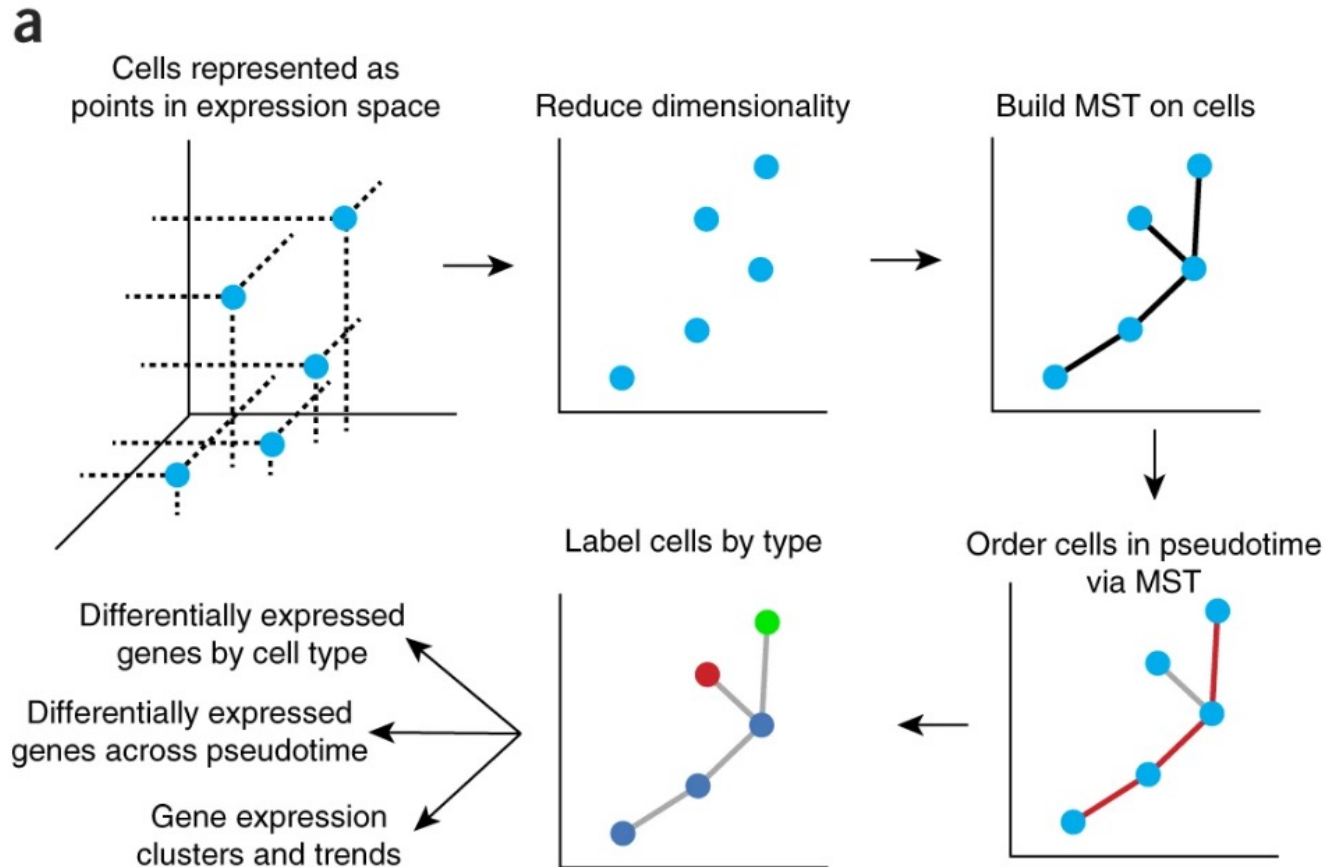
77k Accesses | **2504** Citations | **109** Altmetric | [Metrics](#)

拟时序分析

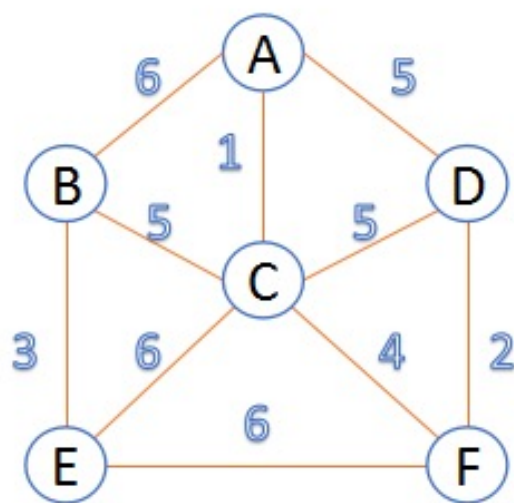
single-cell RNA-Seq data of differentiating myoblasts



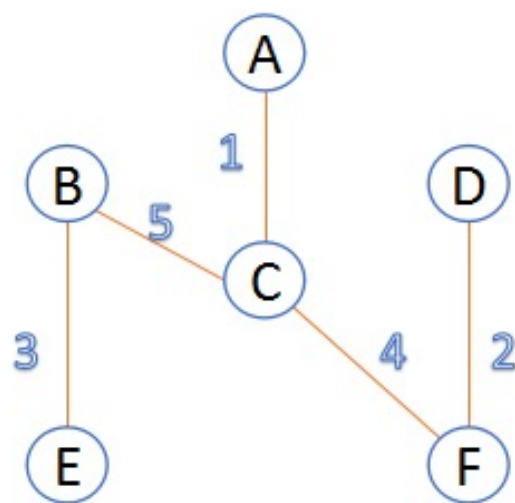
拟时序分析



最小生成树

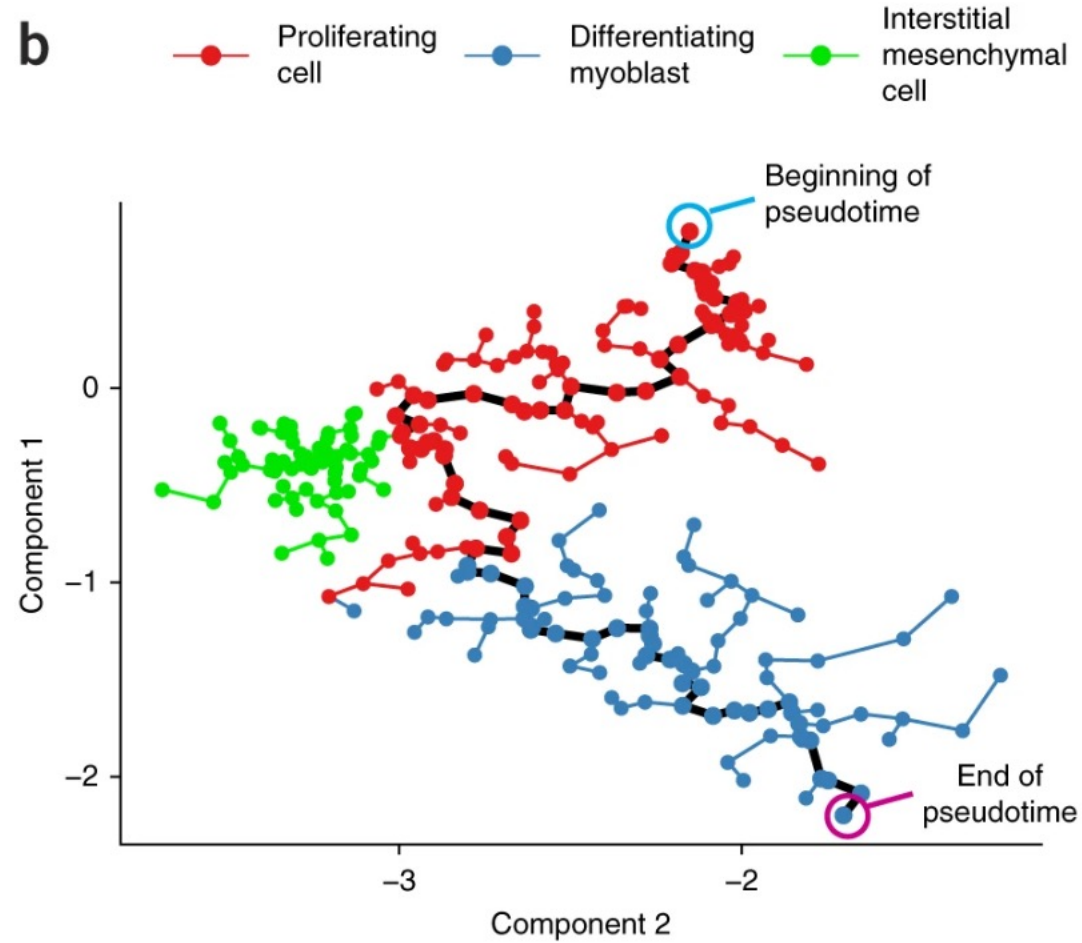


连通网G

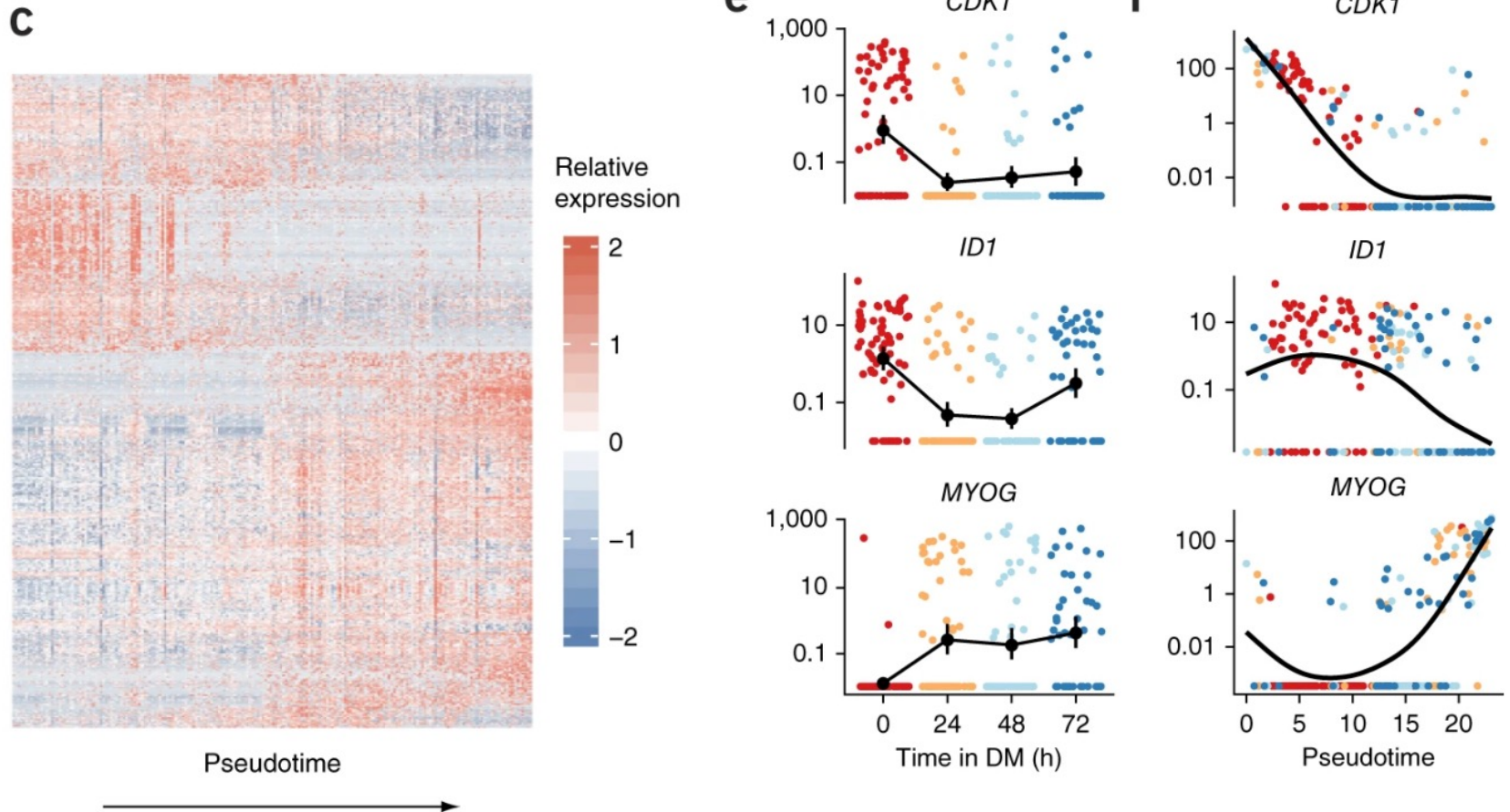


最小生成树

拟时序分析



拟时序分析

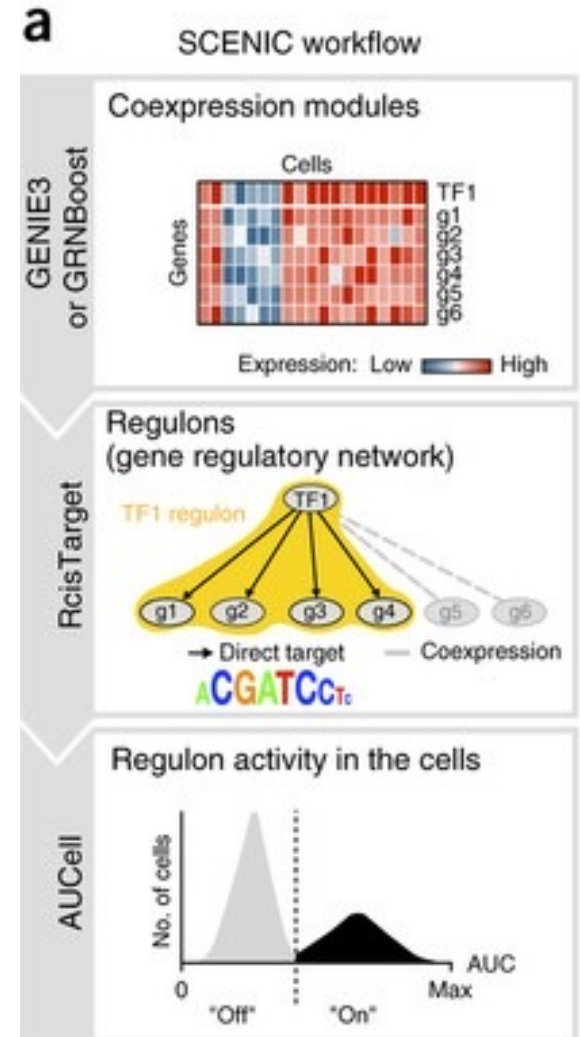


拟时序分析系列方法

Method	SCUBA pseudotime	Wanderlust	Wishbone	SLICER	SCOUP	Waterfall	Mpath	TSCAN	Monocle	SCUBA
Visual abstract										
Structure	Linear	Linear	Single bifurcation	Branching	Branching	Linear	Branching	Linear	Branching	Branching
Robustness strategy	Principal curves	Ensemble, starting cell	Ensemble, starting cell	Starting cell	Starting population	Clustering of cells	Clustering of cells using external labelling	Clustering of cells	Differential expression	Simple model
Extra input requirements	None	Starting cell	Starting cell	Starting cell	Starting population	None	Time points	None	Time points	Time points
Unbiased	+	±	±	±	±	+	-	+	-	-
Scalability w.r.t. cells	-	-	±	±	-	±	+	+	-	±
Scalability w.r.t. genes	+	+	+	+	-	+	±	±	±	+
Code and documentation	-	±	+	±	+	±	+	+	+	±
Parameter ease-of-use	+	+	+	+	-	±	-	+	+	+
First Author	Marco	Bendall	Setty	Welch	Matsumoto	Shin	Chen	Ji	Trapnell	Marco
Last Author	GC Yuan	Dana Pe'er	Dana Pe'er	Hartemink, Prins	Kiryu	Hongjun Song	Poidinger	Ji	Rinn	GC Yuan
Journal	PNAS	Cell	Nature Biotechnology	Genome Biology	BMC Bioinformatics	Cell Stem Cell	Nature Communications	NAR	Nature Biotechnology	PNAS
Year	2014	2014	2016	2016	2016	2015	2016	2016	2014	2014

SCENIC的分析流程

- 共表达模块的构建
 - 与TF共表达的基因归为一个共表达模块
 - 使用GENIE3或GRNBoost算法
- 转录因子调控网络的构建
 - 根据TF的motif筛选共表达模块中的基因，形成最终的网络
 - 网络中每个TF及其靶基因构成一个regulon
 - 使用Rcistarget包
- 转录因子活性评分
 - 使用AUCell对每个regulon进行评分



Regulatory information from single-cell data

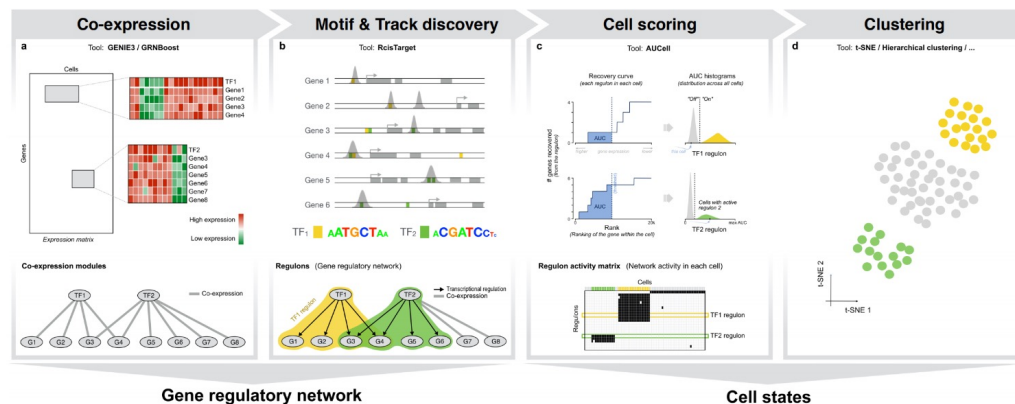
SCENIC Suite: Unveil regulatory information from single-cell data

SCENIC Suite is a set of tools to study and decipher gene regulation. Its core is based on SCENIC (Single-Cell rEgulatory Network Inference and Clustering) which enables you to infer transcription factors, gene regulatory networks and cell types from single-cell RNA-seq data (using SCENIC) or the combination of single-cell RNA-seq and single-cell ATAC-seq data (using SCENIC+).

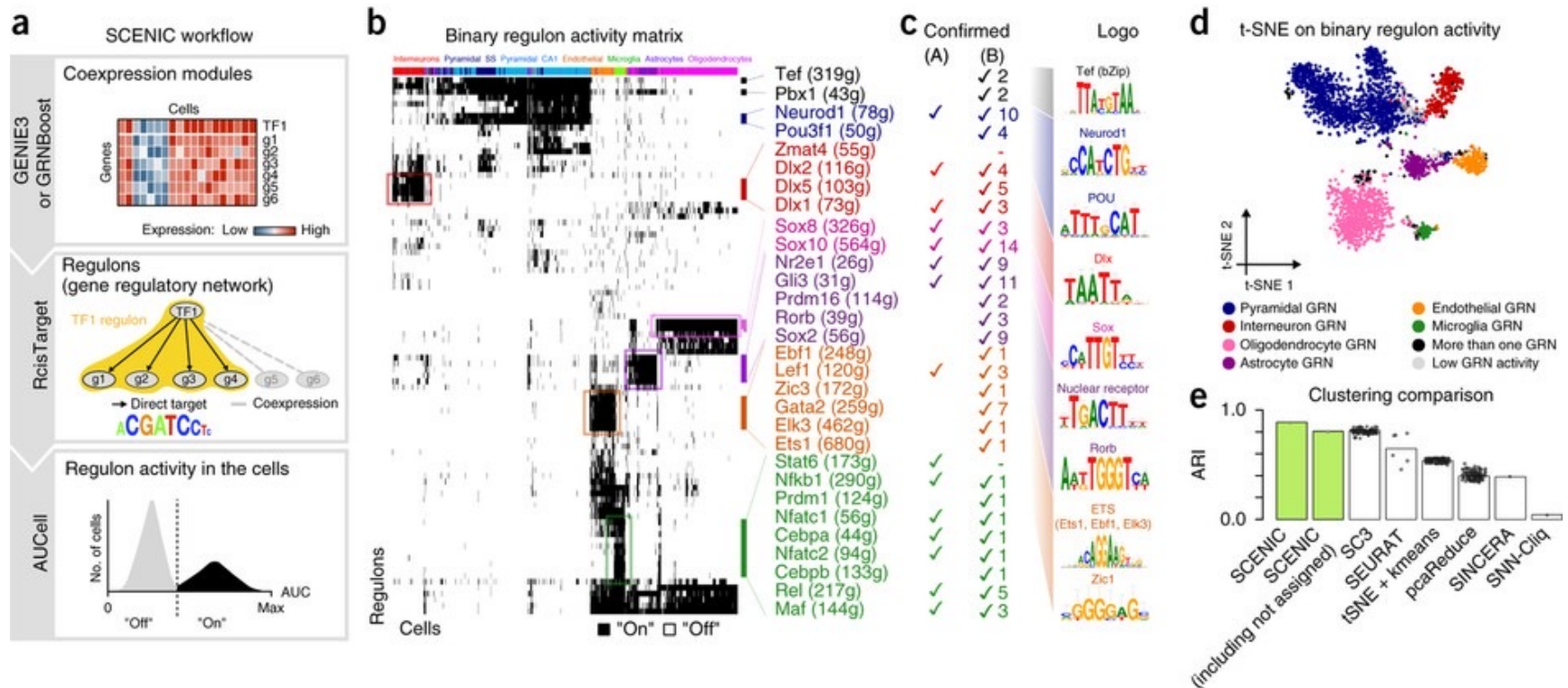
Read the original SCENIC paper (Aibar et al., 2017)

Read the SCENIC protocols paper (Van de Sande et al., 2020)

Read the SCENIC+ preprint (Bravo and De Winter et al., 2022)



The SCENIC workflow and its application to the mouse brain



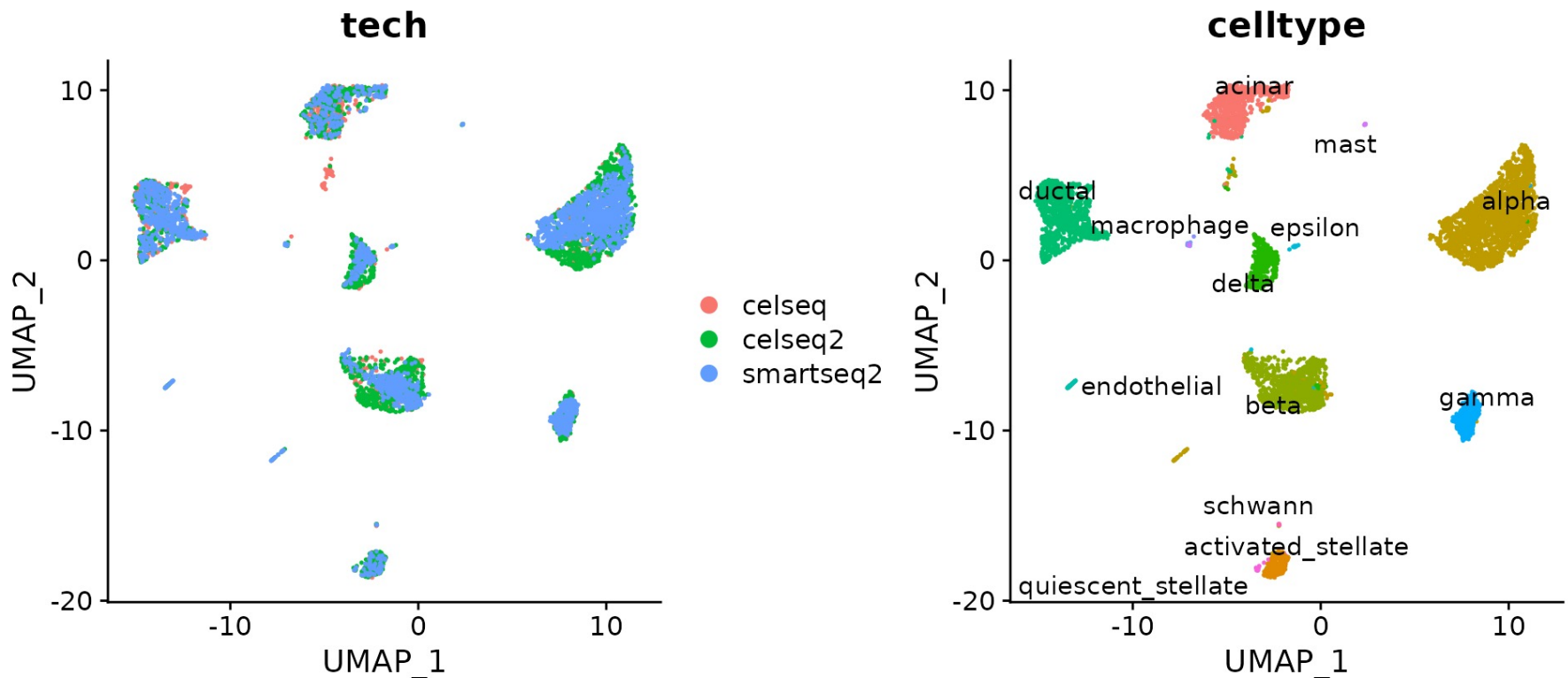
多样本数据整合-批次效应

■ 单细胞测序产生批次效应原因有：

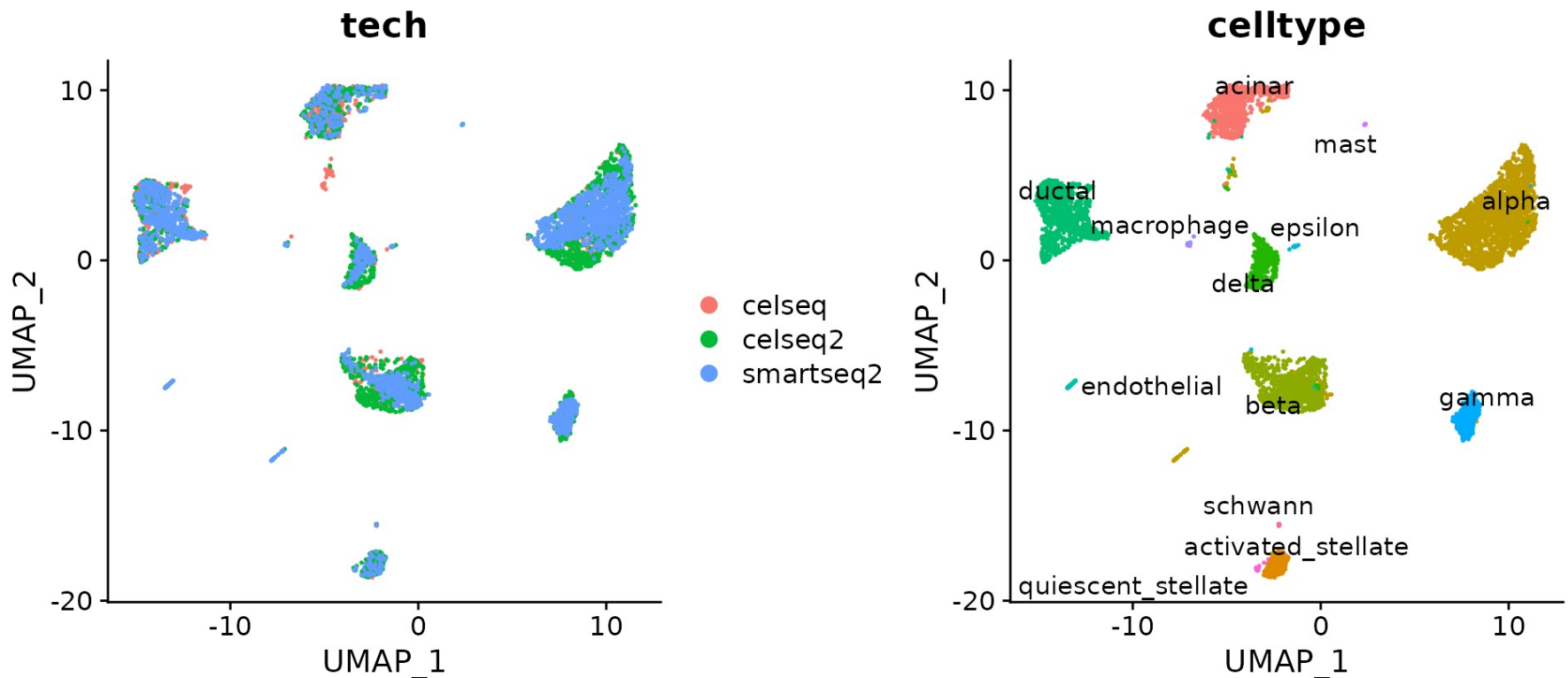
- （1）不同建库策略，如10X平台，Drop-seq，SMART2-seq等
- （2）不同测序平台
- （3）不同公司的试剂或不同批次的试剂
- （4）不同测序批次
- （5）不同实验操作者.....

■ 单细胞数据中最显著的技术变量是计数深度和批次

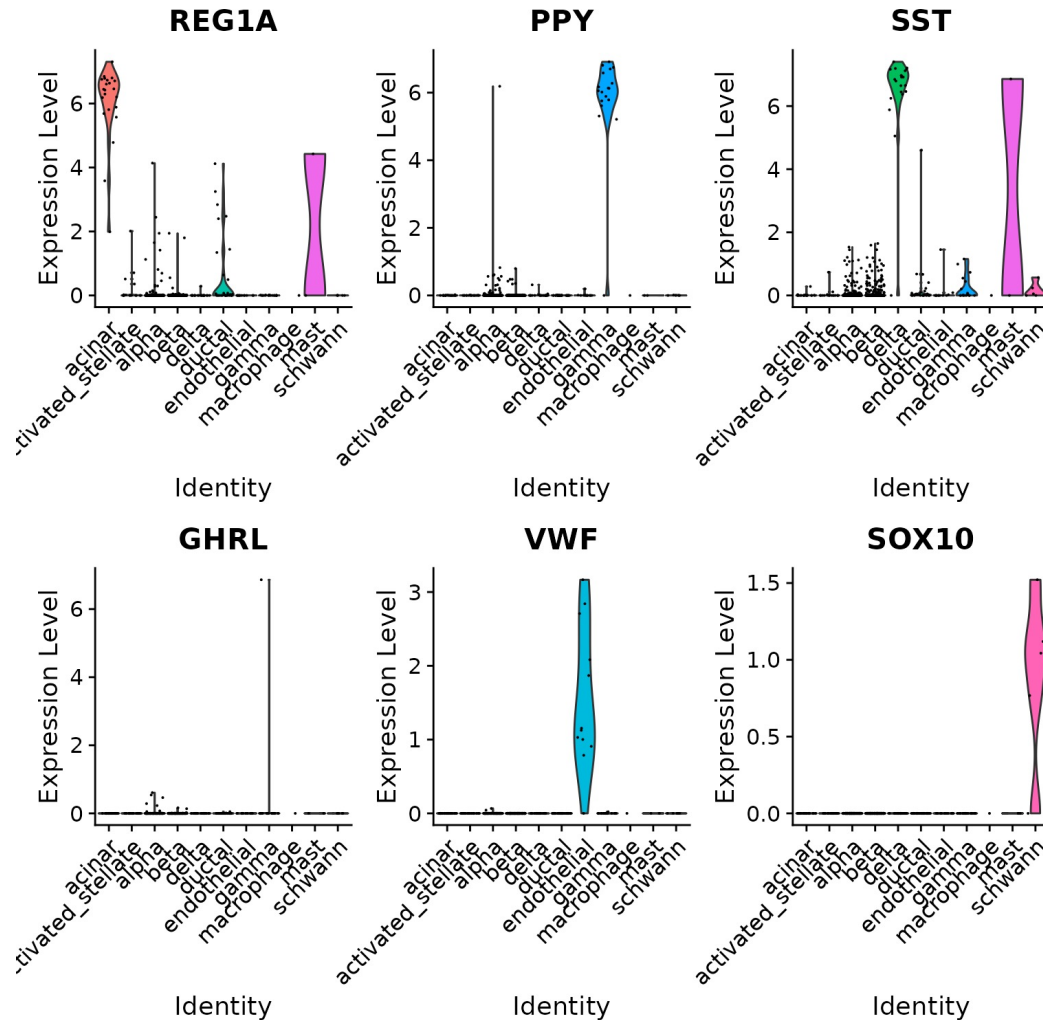
Integration of 3 pancreatic islet cell datasets



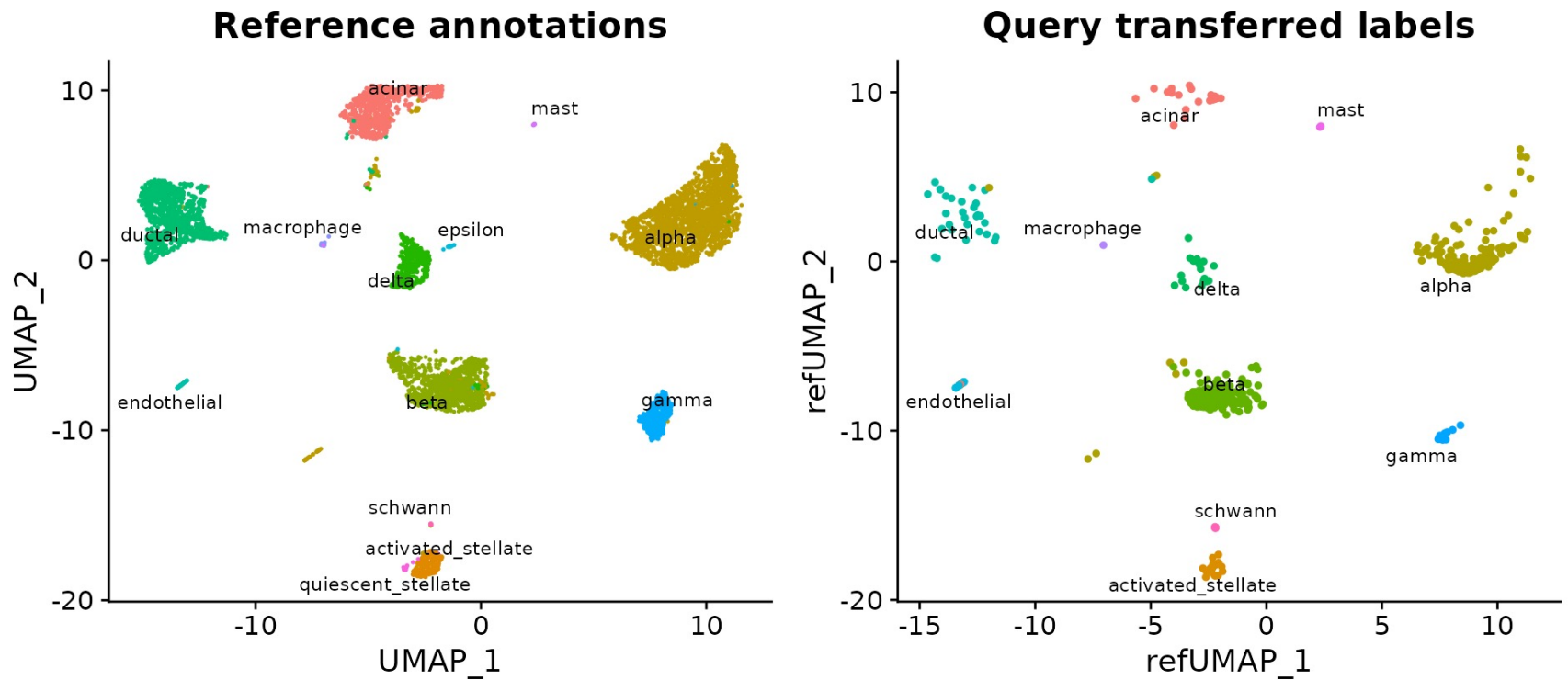
Integration of 3 pancreatic islet cell datasets



Cell type classification using an integrated reference



What is MapQuery doing?



多样本数据整合-常用算法 (Harmony)

■ Harmony方法在2019年提出

■ 优势:

- 整合数据的同时对稀有细胞的敏感性依然很好
- 省内存
- 适合于更复杂的单细胞分析实验设计，可以比较来自不同供体、组织和技术平台的细胞

nature | methods

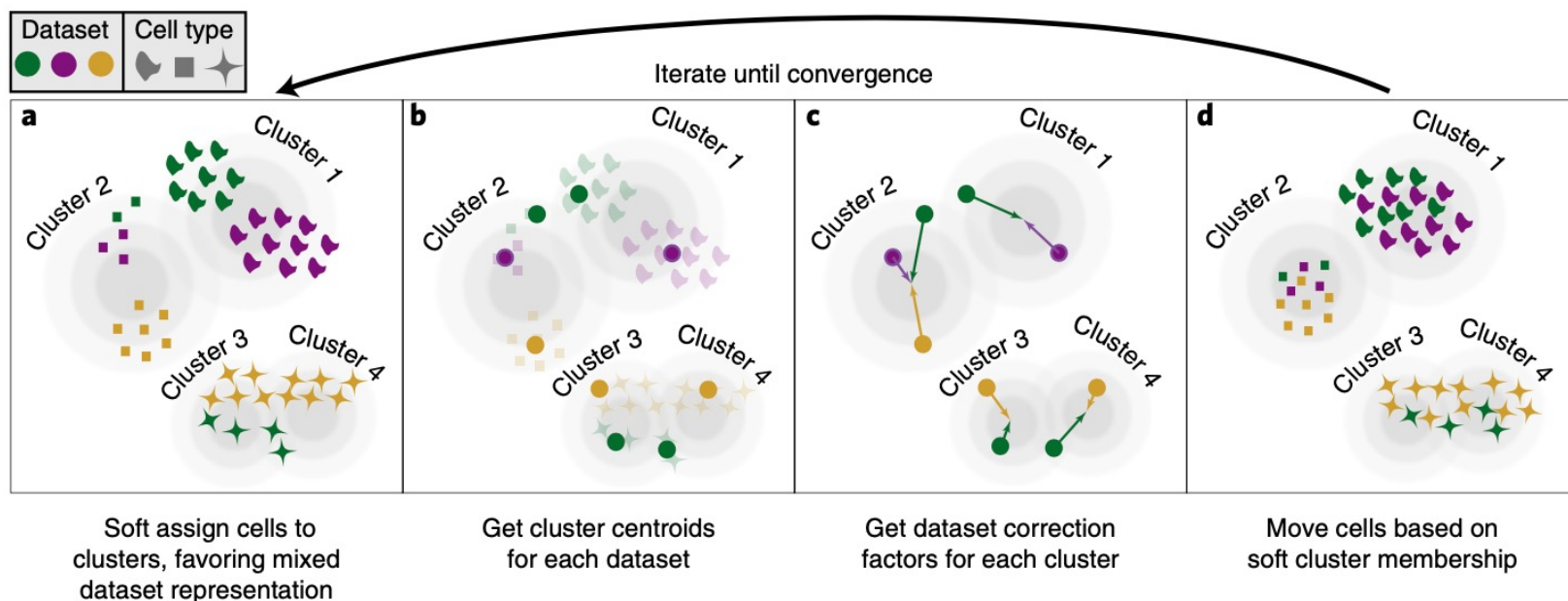
ARTICLES

<https://doi.org/10.1038/s41592-019-0619-0>

Fast, sensitive and accurate integration of single-cell data with Harmony

Ilya Korsunsky^{1,2,3,4}, Nghia Millard^{1,2,3,4}, Jean Fan⁵, Kamil Slowikowski^{1,2,3,4},
Fan Zhang^{1,2,3,4}, Kevin Wei², Yuriy Baglaenko^{1,2,3,4}, Michael Brenner², Po-ru Loh^{1,3,4} and
Soumya Raychaudhuri^{1,2,3,4,6★}

多样本数据整合-常用算法（Harmony）



•A) 使用模糊聚类将每个单元分配到多个聚类，而惩罚项确保每个聚类内数据集的多样性最大化。

•B) 计算每个集群的全局质心，以及每个集群的数据集特定质心。

•C) 在每个集群内，Harmony 根据质心计算每个数据集的校正因子。

•D) 使用特定于细胞的因子校正每个细胞：由在步骤 A 中进行的软集群分配加权的数据集校正因子的线性组合。

Article | [Published: 20 February 2019](#)

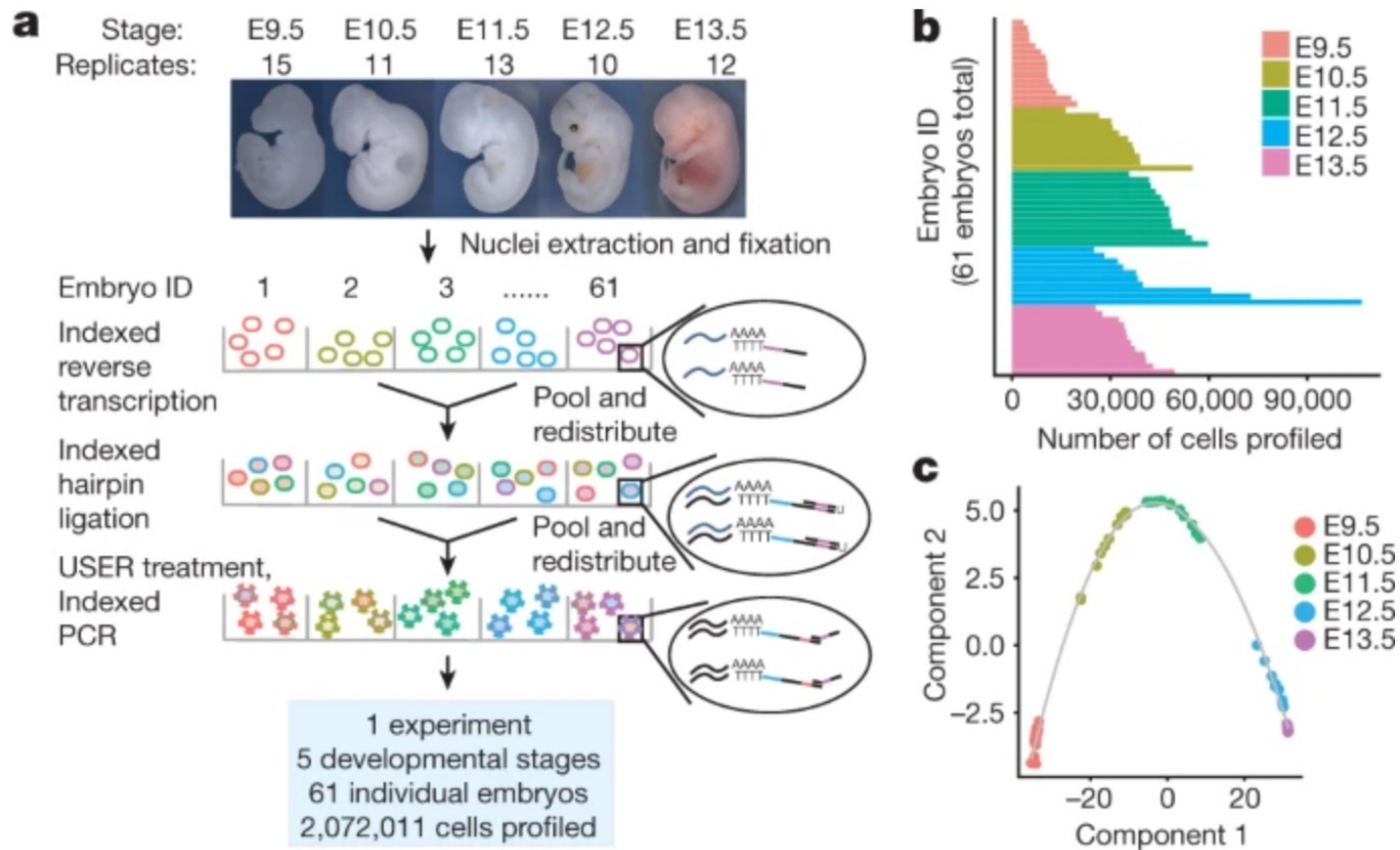
The single-cell transcriptional landscape of mammalian organogenesis

[Junyue Cao](#), [Malte Spielmann](#), [Xiaojie Qiu](#), [Xingfan Huang](#), [Daniel M. Ibrahim](#), [Andrew J. Hill](#), [Fan Zhang](#), [Stefan Mundlos](#), [Lena Christiansen](#), [Frank J. Steemers](#), [Cole Trapnell](#) ✉ & [Jay Shendure](#) ✉

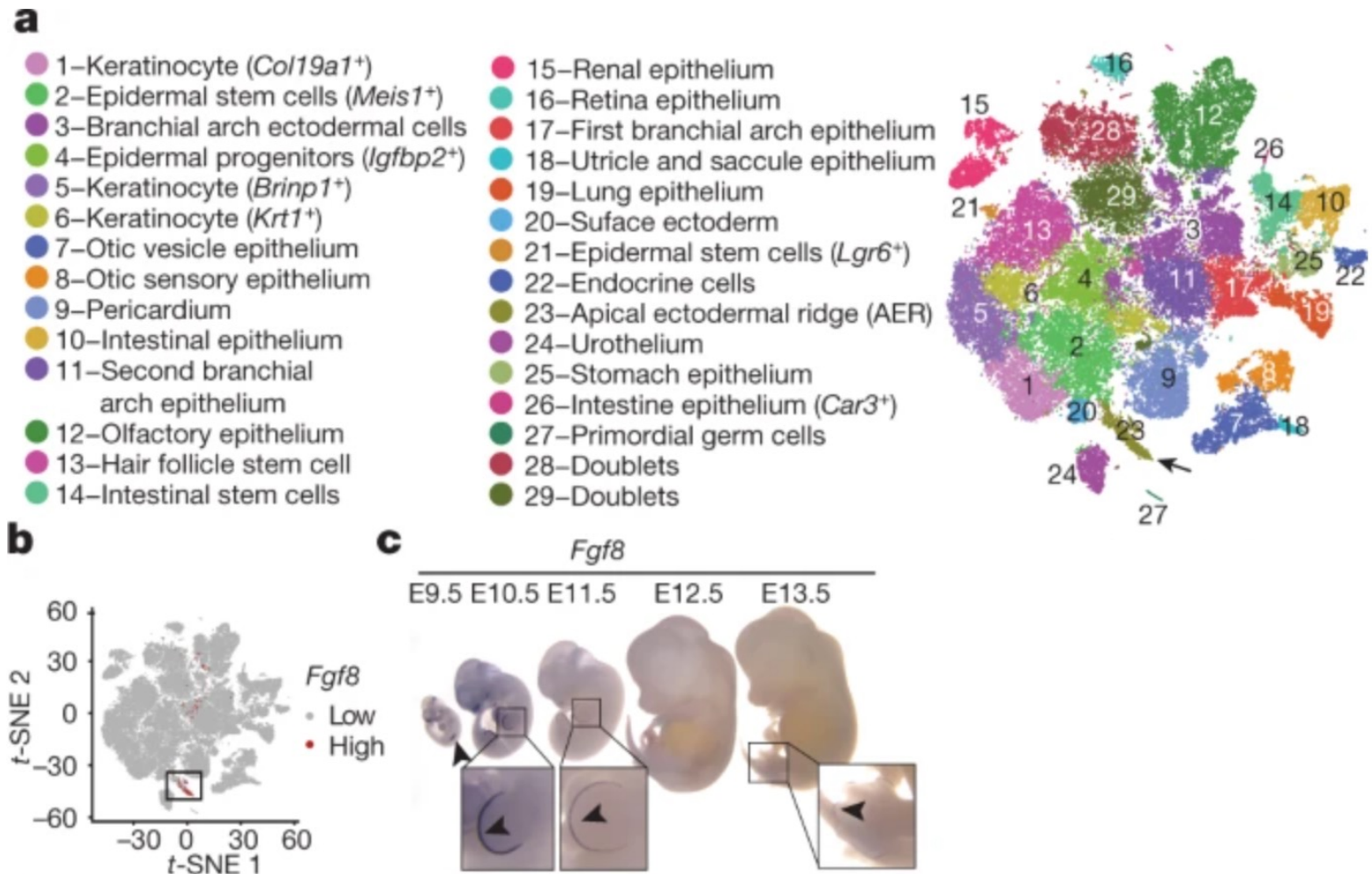
[Nature](#) **566**, 496–502 (2019) | [Cite this article](#)

103k Accesses | **1082** Citations | **501** Altmetric | [Metrics](#)

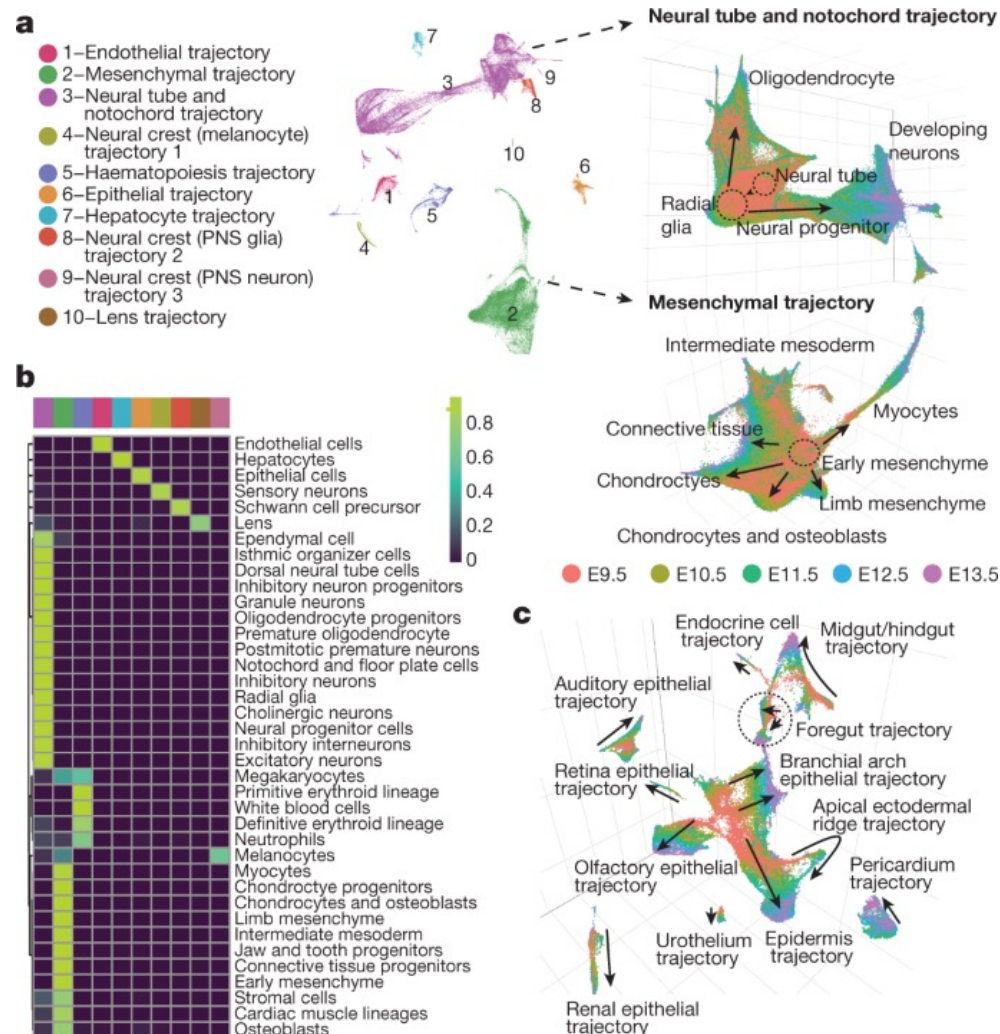
Fig. 1: sci-RNA-seq3 enables profiling of 2,072,011 cells from 61 mouse embryos across 5 developmental stages in a single experiment.



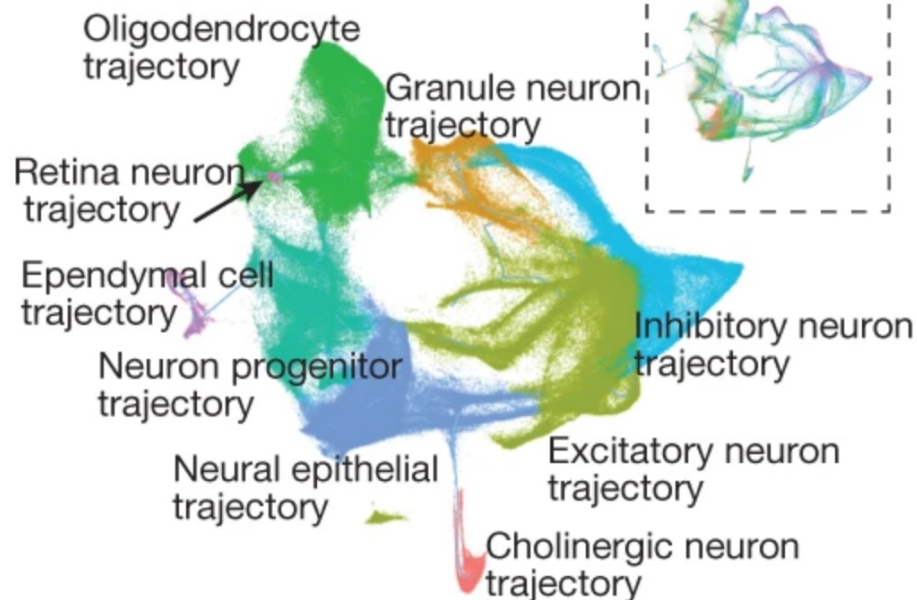
Identification and characterization of epithelial cell subtypes



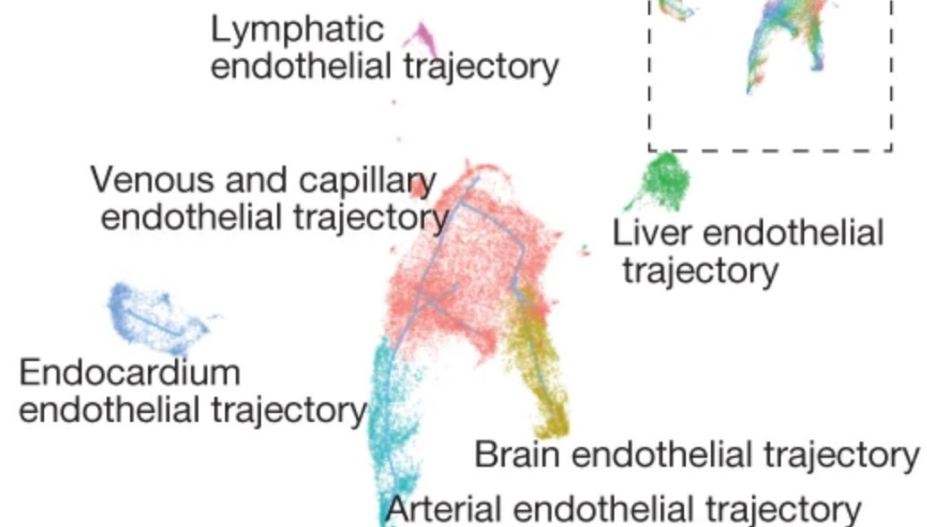
Characterization of ten major developmental trajectories present during mouse organogenesis



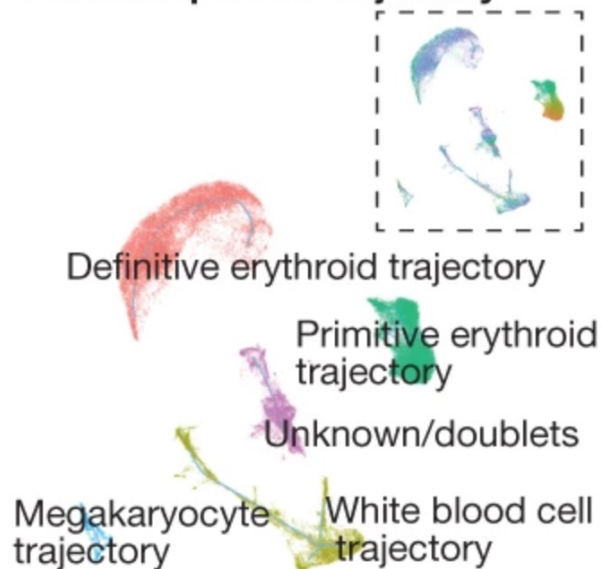
Neural tube/notochord trajectory



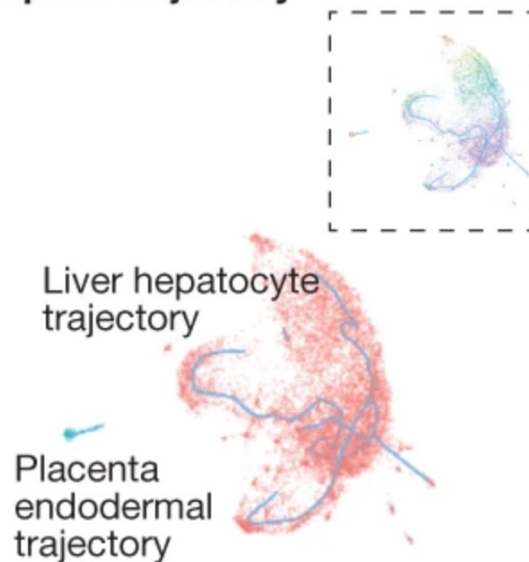
Endothelial trajectory



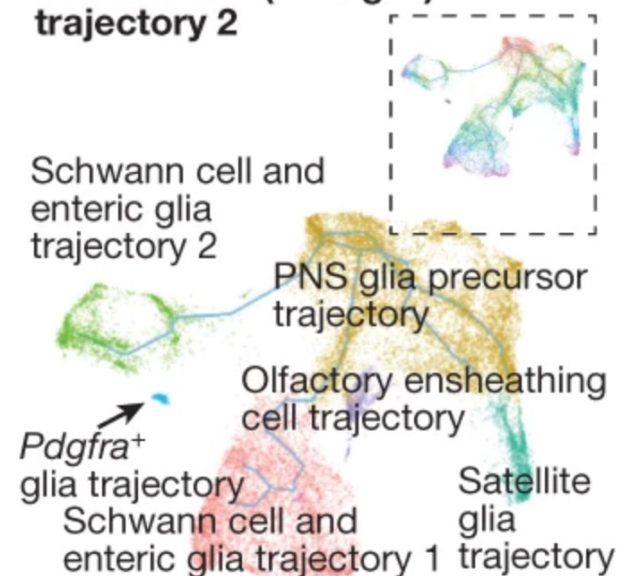
Haematopoiesis trajectory



Hepatic trajectory



Neural crest (PNS glia) trajectory 2



Thanks for attention!
